Discrete Sliding Mode Control with Actuator Constraints for the Stabilization of a Ballistically Launched Drone

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Highly unstable initial states, such as those produced by ballistic deployment, require nonlinear controllers to stabilize the tumble before the vehicle crashes. We derive and test two nonlinear controllers certified through Lyapunov stability analysis: a continuous-time Sliding Mode controller (cSMC) and a discrete-time Sliding Mode Control (dSMC). We test these against two off-the-shelf controllers: the Linear Quadratic Regulator with Integral Action (LQR) and the cascaded Proportional-Integral-Derivative (PID) controller. We find that the dSMC recovers more of the ballistic-deployment envelope than the LQR, PID, or continuous-time controllers, but that enforcing physical motor-speed constraints sharply reduces the recoverable set. We then extend the dSMC to account for possible motor control constraints in terms of maximum squared rotational speed. If the desired motor speeds fall within the physical bounds, then the augmented controller and the ideal controller are identical, and we satisfy the exponential reaching law. Otherwise, we saturate the control inputs in order of priority (first yaw, then altitude/thrust, then pitch/roll), and obtain weaker input-to-state stability (ISS). However, despite this weaker stability, this priority-weighted saturation increases average landing-site reachability roughly fourfold, from 7.9% (naive per-motor clipping) to 32.9% (saturation algorithm).

I. Introduction

In this paper, we develop a controller for UAV stabilization following ballistic deployment. Current work in ballistically launched UAVs, such as the Caltech SQUID [1] or the Texas A&M TLMAV [2], focuses on drones that transition into powered flight at a predetermined point in their ballistic trajectory. Current military research efforts, such as the Launched Effects program [3], call for a flexible platform that can be deployed anywhere, requires minimal infrastructure, and can carry arbitrary payloads. For example, quick-reaction drones could deliver medical supplies, a role examined in feasibility modeling studies [4]. Alternatively, launched drones could create a mobile dispersed sensor network, whether for wide-area surveying [5] or force-protection surveillance [6]. These applications require drones capable of deploying into powered flight at any point along their trajectory. We evaluate the controllers in these test cases by energy efficiency and settling time.

A. Related Work

The dSMC approach for quadrotor control has been the subject of some limited existing studies. Xiong et. al. [7] introduced the unconstrained formulation of the dSMC and presented how a discrete Lyapunov stability condition may be satisfied for arbitrary discretization sampling times. Xu et al. [8] introduced per-channel auxiliary states for the three attitude moments of a quadrotor using SMC under input saturation, and proved closed-loop stability via Lyapunov analysis. Kuang and Chen [9] adopted a vector-valued second-order auxiliary controller for the position and attitude loops together with a disturbance observer and adaptive laws for unknown mass and inertia. Shao et al. [10], on the other hand, embedded a bounded saturation function directly in the control law alongside adaptive parameters. However, these works are all in the continuous time space, and do not cover the collective-thrust control channel together with the three torque channels in a single per-motor construction.

While these saturation methods do not incorporate control prioritization, Faessler et al. [11] present an explicit prioritization of roll and pitch torques above collective thrust above yaw torque under motor saturation, and build the priority into an iterative mixer without formal stability assurances. Similarly, Mueller and D'Andrea [12] linearize their

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input constraints inside an MPC loop. We use the same priority order ($M_x, M_y \succ T \succ M_z$) but solve the projection over the squared rotor speed Ω^2-space and carry it through to the closed-loop stability proof.

In addition to retaining the basic Lyapunov stability bounds, Wang et al. [13] give us the discrete-time component-wise saturation model and the saturation-dependent Lyapunov approach we extend to characterize the admissible control space $\mathcal{U}$, and Jiang and Wang [14] provide the discrete-time input-to-state-stability (ISS) Lyapunov framework we use to demonstrate the ISS form of our closed-loop result. The discrete reaching-law boundary $\eta\,\Delta t < 2$, which follows from the discrete-time sliding-mode existence condition $|s_{k+1}| < |s_k|$ of Sarpturk et al. [15], drops out as a special case when the deficit Δu_k vanishes; the unconstrained dSMC structure follows the discrete-time formulation of Xiong and Zhang [7].

B. New Contributions

We present a corrected derivation with formal Lyapunov stability analysis of the Discrete Sliding Mode Controller (dSMC) presented in [7]. We also extend the dSMC to include per-motor speed limits, which equate to control effort saturation in collective thrust, and roll, pitch, and yaw moments. We compare this to a reformulated Continuous Sliding Mode Controller (cSMC), similar to the form derived in [16]. We compare these two SMC formulations against classical LQR and cascaded PID, both of which are common commercial off-the-shelf (COTS) drone control solutions. Our dSMC corrects and re-derives the control laws in [7], whose proposed forms violate the exponential reaching condition and produce an unstable or unresponsive controller. We also introduce a new test case, the stabilization of a ballistically launched quadrotor into powered flight, which suggests that the results of [16] hold only for a limited range of initial conditions. Lastly, we reformulate the dSMC controller with control saturation. We first apply control-priority weighting (with pitch and roll weighted as more important than total thrust, which in turn is weighted as more important than yaw). Then, we augment each control channel with auxiliary states and, once again, apply the exponential reaching law analysis, demonstrating that the reaching laws hold under transient control saturation. When commands are naturally feasible, we recover the unconstrained dSMC controller. In the presence of control saturation, we demonstrate an input-to-state stability bound (provided the saturation is not permanent). This approach is similar to [13, 14], but we adapt these methods to discrete time control with a separate scalar auxiliary state per actuator channel (including total thrust) and show that we still maintain the Lyapunov bound of the underlying dSMC under transient saturation. We also examine a new application of the controller; ballistic transition stabilization and landing.

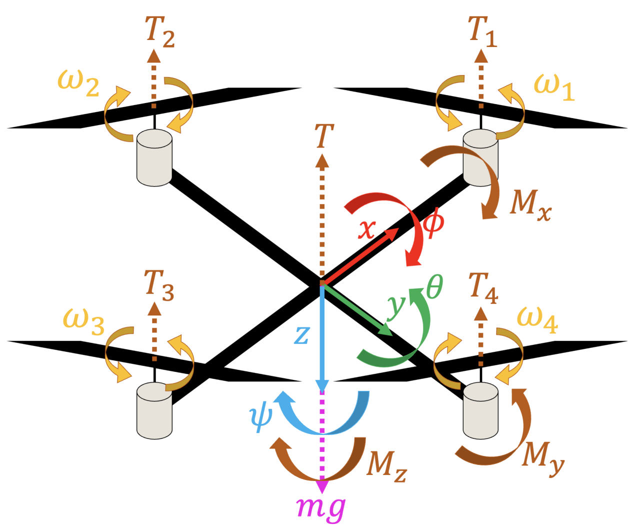

Fig. 1 Quadrotor coordinate system and force vectors.

II. Vehicle Model

We apply the controllers derived in this paper to the standard symmetric quadrotor in a North-East-Down inertial coordinate frame with Z-Y-X Euler angle rotation order. We use the standard nonlinear and linear system of equations presented in [16]. The model parameters are given in Table 1. We choose to work in the force-moment space (T for

Table 1 Nomenclature and Variables

Symbol	Description	Value
Model parameters		
k_{MT}	Ratio of motor torque to thrust	0.1 m
l	Quadrotor arm length	0.2 m
J_{xx}	Moment of inertia about body x axis	1.8×10^{-3} kg · m^2
J_{yy}	Moment of inertia about body y axis	1.8×10^{-3} kg · m^2
J_{zz}	Moment of inertia about body z axis	1.5×10^{-3} kg · m^2
m	Drone mass	0.8 kg
g	Acceleration due to gravity	9.81 m/s^2
J_r	Rotor moment of inertia about the spin axis of the motor	≈ 0
κ	Gyroscopic coupling between rotors and body	≈ 0 (see text)
State variables		
v_x, v_y, v_z	Body-frame velocity components	–
$\omega_x, \omega_y, \omega_z$	Body-frame rotation rates	–
θ, ϕ, ψ	Pitch, roll, and yaw Euler angles (Z-Y-X order)	–
x, y, z	Inertial position; north, east positive; z upward (altitude)	–
Motor mixing		
T	Net thrust along the body z axis	–
M_x, M_y, M_z	Moments about the body x, y, z axes	–
T_i	Thrust produced by rotor i, $i \in \{1, \ldots, 4\}$	–
ω_i	Angular velocity of rotor i	–
Ω_i^2	Squared angular velocity of rotor i	–
Ω_r	Net rotor speed differential $\omega_1 - \omega_2 + \omega_3 - \omega_4$	–
$\Omega^2_{\text{min}}, \Omega^2_{\text{max}}$	Per-rotor squared-speed bounds	0, 2 (normalized)
T_M	Mixing matrix from Ω^2 to $(T, M_x, M_y, M_z)^T$	–
b	Per-rotor thrust per squared rotor speed	5 (normalized)
d	Per-rotor yaw torque per squared rotor speed	2 (normalized)
u_k	Commanded control at step k, $(T, M_x, M_y, M_z)^T$	–
$\bar{u}_k$	Saturated command after projection onto $\mathcal{U}$	–
Δu_k	Saturation deficit, $u_k - \bar{u}_k$	–
$\mathcal{U}$	Feasible control set under per-rotor speed limits	–
Sliding mode control and saturation		
s_α	Sliding surface for channel $\alpha \in \{T, M_x, M_y, M_z\}$	–
$\tilde{s}_k$	Modified sliding surface, $s_k - D_k\,\xi_k$	–
e_α	Tracking error for channel α	–
η_α	Reaching-law gain for channel α	–
ξ_k	Per-channel auxiliary state vector	–
$k_{\xi,\alpha}$	Contraction rate of the auxiliary-state filter	–
D_k	Coupling between the auxiliary state and the sliding surface	–
V_k	Discrete Lyapunov function at step k	–
Model discretization		
Δt	Discrete control sample interval	0.02 s
k	Discrete-time index	–

net thrust along the body z axis and M_x, M_y, M_z for the moments about the body x, y, and z axes), rather than the individual rotor angular velocities $(\omega_1, \omega_2, \omega_3, \omega_4)$, without loss of generality due to the linear mapping from squared rotor velocities to force-moments.

$$\begin{bmatrix} T \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ k_{MT} & -k_{MT} & k_{MT} & -k_{MT} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}. \tag{1}$$

The full nonlinear system of equations of motion is given in Equation 2, where v_x, v_y, v_z are the body-frame velocity components, $\omega_x, \omega_y, \omega_z$ are the body-frame attitude rates, θ, ϕ, ψ are the inertial pitch, roll, and yaw Euler angles, and x, y, z are the inertial position (north and east positive; the z position is expressed as altitude, positive upward).

$$\begin{aligned}
\dot{v}_x &= -v_z\,\omega_y + v_y\,\omega_z - g\ \mathrm{s}\,\theta, & \dot{v}_y &= -v_x\,\omega_z + v_z\,\omega_x + g\ \mathrm{c}\,\theta\ \mathrm{s}\,\phi, & \dot{v}_z &= -v_y\,\omega_x + v_x\,\omega_y + g\ \mathrm{c}\,\theta\ \mathrm{c}\,\phi - \tfrac{T}{m},\\
\dot{\omega}_x &= \tfrac{1}{J_{xx}}\big[-\omega_y\,\omega_z(J_{zz}-J_{yy}) & \dot{\omega}_y &= \tfrac{1}{J_{yy}}\big[-\omega_x\,\omega_z(J_{xx}-J_{zz}) & \dot{\omega}_z &= \tfrac{M_z}{J_{zz}},\\
&\quad + M_x - \kappa\,M_z\,\omega_y\big], & &\quad + M_y - \kappa\,M_z\,\omega_x\big], & &\\
\dot{\theta} &= \omega_y\ \mathrm{c}\,\phi - \omega_z\ \mathrm{s}\,\phi, & \dot{\phi} &= \omega_x + \omega_y\ \mathrm{s}\,\phi\ \mathrm{t}\,\theta + \omega_z\ \mathrm{c}\,\phi\ \mathrm{t}\,\theta, & \dot{\psi} &= \tfrac{\omega_y\ \mathrm{s}\,\phi + \omega_z\ \mathrm{c}\,\phi}{\mathrm{c}\,\theta},\\
\dot{x} &= v_x\ \mathrm{c}\,\psi\ \mathrm{c}\,\theta & \dot{y} &= v_x\ \mathrm{s}\,\psi\ \mathrm{c}\,\theta & \dot{z} &= v_x\ \mathrm{s}\,\theta - v_y\ \mathrm{c}\,\theta\ \mathrm{s}\,\phi - v_z\ \mathrm{c}\,\theta\ \mathrm{c}\,\phi.\\
&\quad + v_y(-\mathrm{s}\,\psi\ \mathrm{c}\,\phi + \mathrm{c}\,\psi\ \mathrm{s}\,\theta\ \mathrm{s}\,\phi) & &\quad + v_y(\mathrm{c}\,\psi\ \mathrm{c}\,\phi + \mathrm{s}\,\psi\ \mathrm{s}\,\theta\ \mathrm{s}\,\phi) & &\\
&\quad + v_z(\mathrm{s}\,\psi\ \mathrm{s}\,\phi + \mathrm{c}\,\psi\ \mathrm{s}\,\theta\ \mathrm{c}\,\phi), & &\quad + v_z(-\mathrm{c}\,\psi\ \mathrm{s}\,\phi + \mathrm{s}\,\psi\ \mathrm{s}\,\theta\ \mathrm{c}\,\phi), & &
\end{aligned} \tag{2}$$

In the simulations reported here the rotor–body gyroscopic terms are inactive: the rotor spin inertia is negligible ($J_r \approx 0$), and the $\kappa\,M_z\,\omega_x$, $\kappa\,M_z\,\omega_y$ coupling terms are disabled accordingly ($\kappa \to 0$); both are retained in the equations above for generality.

Linearizing the system about the steady-state point (level and stationary) yields a simplified set of equations that we use to derive our LQR controller gains.

$$\begin{aligned}
\dot{v}_x &= -g\,\theta, & \dot{v}_y &= g\,\phi, & \dot{v}_z &= -T/m,\\
\dot{\omega}_x &= M_x/J_{xx}, & \dot{\omega}_y &= M_y/J_{yy}, & \dot{\omega}_z &= M_z/J_{zz},\\
\dot{\theta} &= \omega_y, & \dot{\phi} &= \omega_x, & \dot{\psi} &= \omega_z,\\
\dot{x} &= v_x, & \dot{y} &= v_y, & \dot{z} &= -v_z.
\end{aligned} \tag{3}$$

We skip the controllability and observability analysis. The quadrotor is well known to be fully (directly or indirectly) controllable in the nonlinear form above. We assume the full state is observable; on our hardware target, GPS, an onboard IMU, and an EKF provide position, attitude, and translational and rotational velocities with sufficient precision.

III. Controller Formulation

A. LQR Controller with Integral Action

LQR computes a constant state-feedback gain that minimizes an infinite-horizon quadratic cost. The cost function we minimize, given in Equation 4, is the standard form of the LQR cost function with integral action.

$$\begin{aligned}
J(u) &= \int_0^\infty \left(\tilde{x}^\top Q\,\tilde{x} + u^\top R\,u\right) \mathrm{d}t\\
\tilde{x} &= \left[\left(\int_0^t e(\tau)\,\mathrm{d}\tau\right)^\top \quad x^\top\right]^\top\\
e &= r - [x, y, z]^\top, \qquad u = \frac{K_i}{s}\,e - K_p\,x
\end{aligned} \tag{4}$$

To realize a LQR controller with integral action, we augment our base state vector with three additional states, representing integrated Earth-frame position. This way, we can derive integral gains in addition to proportional gains

Table 2 PID gains for outer/inner position and attitude loops.

Loop	Channel	P	I	D	N
Outer positional	$x \to \dot{x}$	0.25	0	0	–
	$y \to \dot{y}$	0.25	0	0	–
	$z \to \dot{z}$	0.40	0	0	–
Inner positional	$\dot{x} \to \ddot{x}$	0.30	0	0.20	100
	$\dot{y} \to \ddot{y}$	0.30	0	0.20	100
	$\dot{z} \to \ddot{z}$	0.80	0	0	–
Outer attitude	$\theta \to \dot{\theta}$	0.30	0	0	–
	$\phi \to \dot{\phi}$	0.30	0	0	–
	$\psi \to \dot{\psi}$	0.30	0	0	–
Inner attitude	$\dot{\theta} \to \ddot{\theta}$	0.15	0	0	–
	$\dot{\phi} \to \ddot{\phi}$	0.15	0	0	–
	$\dot{\psi} \to \ddot{\psi}$	0.15	0	0	–

Table 3 PD controller gains for outer/inner position loops.

Loop	Channel	P	D	N
Outer positional	$x \to \dot{x}$	1	6	50
	$y \to \dot{y}$	1	6	50
	$z \to \dot{z}$	2	0	–
Inner positional	$\dot{x} \to \ddot{x}$	1	1	50
	$\dot{y} \to \ddot{y}$	1	1	50
	$\dot{z} \to \ddot{z}$	1	0	–

when we solve the Discrete Algebraic Riccati Equation. We use the following state-cost matrix Q and control-cost matrix R, which are the result of manual tuning of the weights presented in [16]:

$$
\begin{aligned}
&Q = \operatorname{diag}(q) \big/ \left|q\right|_2, \quad R = I_4, \\
&q = [3,\ 3,\ 6000,\ 1080,\ 1080,\ 1080,\ 180,\ 180,\ 180,\ 0.5,\ 0.5,\ 1000,\ 15,\ 15,\ 300]^\top.
\end{aligned} \tag{5}
$$

Solving for the optimal proportional and integral gains, we obtain:

$$
\begin{aligned}
K_i &= \begin{bmatrix} 0 & 0 & 0.22 \\ 0 & 0.05 & 0 \\ -0.05 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
K_p &= \begin{bmatrix} 0 & 0 & -1.55 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.91 \\ 0 & 0.16 & 0 & 0.42 & 0 & 0 & 0 & 1.56 & 0 & 0 & 0.13 & 0 \\ -0.16 & 0 & 0 & 0 & 0.42 & 0 & 1.16 & 0 & 0 & -0.13 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.41 & 0 & 0 & 0.17 & 0 & 0 & 0 \end{bmatrix}
\end{aligned} \tag{6}
$$

B. Cascaded PID Controller

We formulate the standard four-loop cascaded PID controller, with outer and inner loops for both position and attitude. The gains used are given in Table 2.

C. Continuous Sliding Mode Controller

The cSMC is a Lyapunov-stable controller that directly accounts for the system's nonlinear dynamics. We follow the control laws of [16] with one modification. Where [16] uses a single PD layer to produce the desired Euler angles, we use two discrete PID layers: the first computes a desired translational acceleration, and the second resolves the corresponding attitude reference. This uses the cascaded structure used by most COTS flight controllers, and lets us tune the per-axis gains independently. The position PD control layer gains are listed in Table 3.

$$\alpha \in \{T, M_x, M_y, M_z\}, \qquad s_\alpha = c_\alpha\, e_\alpha + \dot{e}_\alpha,$$

$$r_{PID} = \begin{bmatrix} \tilde{z} \\ \tilde{\phi} \\ \tilde{\theta} \\ \tilde{\psi} \end{bmatrix}, \quad e = r_{PID} - \begin{bmatrix} z \\ \phi \\ \theta \\ \psi \end{bmatrix}, \quad \begin{bmatrix} c_T \\ c_{M_x} \\ c_{M_y} \\ c_{M_z} \end{bmatrix} = \begin{bmatrix} 0.5 \\ 4 \\ 8 \\ 0.5 \end{bmatrix}. \tag{7}$$

$$\begin{aligned} \hat{u}_T &= \frac{m\, c_T}{\mathrm{c}\,\theta\ \mathrm{c}\,\phi}\big(v_x\ \mathrm{s}\,\theta - v_y\ \mathrm{s}\,\theta\ \mathrm{s}\,\phi - v_z\ \mathrm{c}\,\theta\ \mathrm{c}\,\phi\big) \\ &\quad + m\big(-v_y\,\omega_x + v_x\,\omega_y + g\ \mathrm{c}\,\theta\ \mathrm{c}\,\phi\big) \end{aligned} \tag{8}$$

$$\hat{u}_{M_x} = -c_{M_x}\, J_{xx}\big(\omega_x + \omega_y\ \mathrm{s}\,\phi\ \mathrm{t}\,\theta + \omega_z\ \mathrm{c}\,\phi\ \mathrm{t}\,\theta\big) + \omega_y\,\omega_z\,(J_{zz} - J_{yy}) \tag{9}$$

$$\hat{u}_{M_y} = -\frac{c_{M_y}\, J_{zz}}{\mathrm{c}\,\phi}\big(\omega_y\ \mathrm{c}\,\phi - \omega_z\ \mathrm{s}\,\phi\big) + \omega_x\,\omega_z\,(J_{xx} - J_{zz}) \tag{10}$$

$$\hat{u}_{M_z} = -c_{M_z}\, J_{zz}\big(\omega_y\ \mathrm{t}\,\phi + \omega_z\big) \tag{11}$$

$$u_\alpha = \hat{u}_\alpha - K_\alpha\ \mathrm{sat}(2 s_\alpha, -1, 1) \tag{12}$$

$$\begin{aligned} K_T &= \frac{\gamma_T\, m}{\mathrm{c}\,\theta\ \mathrm{c}\,\phi}, \quad K_{M_x} = \gamma_{M_x}\, J_{xx}, \\ K_{M_y} &= \frac{\gamma_{M_y}\, J_{yy}}{\mathrm{c}\,\phi}, \quad K_{M_z} = \frac{\gamma_{M_z}\, J_{zz}\ \mathrm{c}\,\theta}{\mathrm{c}\,\phi}, \\ \gamma_T &= 12, \quad \gamma_{M_x} = 8, \quad \gamma_{M_y} = 8, \quad \gamma_{M_z} = 1. \end{aligned} \tag{13}$$

We now present a brief Lyapunov stability proof for the cSMC formulation. We define the Lyapunov function $V_\alpha(x)$ and its time derivative for each control channel $\alpha \in \{T, M_x, M_y, M_z\}$:

$$V_\alpha(x) = \tfrac{1}{2}\, s_\alpha^2, \qquad \dot{V}_\alpha(x) = s_\alpha\, \dot{s}_\alpha. \tag{14}$$

The goal is to show that if s_α stably converges to zero, the tracking error of the controller likewise converges to zero. Since $\dot{s}_\alpha$ is the error response of the system (from Equation 7), we derive $\dot{s}_\alpha$ directly from the nonlinear equations of motion (Equation 2):

$$\begin{aligned} \dot{s}_T &= c_T\big(v_x\,\mathrm{s}\,\theta - v_y\,\mathrm{s}\,\theta\,\mathrm{s}\,\phi - v_z\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi\big) \\ &\quad - \left(-v_y\,\omega_x + v_x\,\omega_y + g\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi - \tfrac{u_T}{m}\right)\ \mathrm{c}\,\theta\,\mathrm{c}\,\phi, \\ \dot{s}_{M_x} &= c_{M_x}\big(\omega_x + \omega_y\,\mathrm{s}\,\phi\,\mathrm{t}\,\theta + \omega_z\,\mathrm{c}\,\phi\,\mathrm{t}\,\theta\big) \\ &\quad - \frac{1}{J_{xx}}\big(\omega_y\omega_z(J_{zz} - J_{yy}) - u_{M_x}\big), \\ \dot{s}_{M_y} &= c_{M_y}\big(\omega_y\,\mathrm{c}\,\phi - \omega_z\,\mathrm{s}\,\phi\big) - \frac{\mathrm{c}\,\phi}{J_{yy}}\big(\omega_x\omega_z(J_{xx} - J_{zz}) - u_{M_y}\big), \\ \dot{s}_{M_z} &= c_{M_z}\left(\frac{\omega_y\,\mathrm{s}\,\phi}{\mathrm{c}\,\theta} + \frac{\omega_z\,\mathrm{c}\,\phi}{\mathrm{c}\,\theta}\right) + \frac{u_{M_z}}{J_{zz}}\ \frac{\mathrm{c}\,\phi}{\mathrm{c}\,\theta}. \end{aligned} \tag{15}$$

Substituting the control efforts from Equations 8–13, with K_α given in their γ_α form:

$$
\begin{aligned}
\dot{s}_T &= c_T\big(v_x\,\mathrm{s}\,\theta - v_y\,\mathrm{s}\,\theta\,\mathrm{s}\,\phi - v_z\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi\big) \\
&\quad - \Bigg[\big(-v_y\,\omega_x + v_x\,\omega_y + g\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi\big) + \frac{1}{m}\Bigg(\frac{mc_T}{\mathrm{c}\,\theta\,\mathrm{c}\,\phi}\big(v_x\,\mathrm{s}\,\theta - v_y\,\mathrm{s}\,\theta\,\mathrm{s}\,\phi - v_z\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi\big) \\
&\qquad + m\big(-v_y\,\omega_x + v_x\,\omega_y + g\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi\big) + \frac{\gamma_T m}{\mathrm{c}\,\theta\,\mathrm{c}\,\phi}\ \mathrm{sat}(2s_T, -1, 1)\Bigg)\Bigg]\,\mathrm{c}\,\theta\,\mathrm{c}\,\phi, \\
\dot{s}_{M_x} &= c_{M_x}\big(\omega_x + \omega_y\,\mathrm{s}\,\phi\ \mathrm{t}\,\theta + \omega_z\,\mathrm{c}\,\phi\ \mathrm{t}\,\theta\big) \\
&\quad - \frac{1}{J_{xx}}\Big[\omega_y\omega_z(J_{zz} - J_{yy}) + \big(-c_{M_x}J_{xx}\big(\omega_x + \omega_y\,\mathrm{s}\,\phi\ \mathrm{t}\,\theta + \omega_z\,\mathrm{c}\,\phi\ \mathrm{t}\,\theta\big) \\
&\qquad + \omega_y\omega_z(J_{zz} - J_{yy}) + \gamma_{M_x}J_{xx}\ \mathrm{sat}(2s_{M_x}, -1, 1)\big)\Big], \\
\dot{s}_{M_y} &= c_{M_y}\big(\omega_y\,\mathrm{c}\,\phi - \omega_z\,\mathrm{s}\,\phi\big) \\
&\quad - \frac{\mathrm{c}\,\phi}{J_{yy}}\Bigg[\omega_x\omega_z(J_{xx} - J_{zz}) + \Bigg(-\frac{c_{M_y}J_{zz}}{\mathrm{c}\,\phi}\big(\omega_y\,\mathrm{c}\,\phi - \omega_z\,\mathrm{s}\,\phi\big) + \omega_x\omega_z(J_{xx} - J_{zz}) \\
&\qquad + \frac{\gamma_{M_y}J_{yy}}{\mathrm{c}\,\phi}\ \mathrm{sat}(2s_{M_y}, -1, 1)\Bigg)\Bigg], \\
\dot{s}_{M_z} &= c_{M_z}\left(\frac{\omega_y\,\mathrm{s}\,\phi}{\mathrm{c}\,\theta} + \frac{\omega_z\,\mathrm{c}\,\phi}{\mathrm{c}\,\theta}\right) \\
&\quad + \left(\frac{-c_{M_z}J_{zz}\big(\omega_y\,\mathrm{t}\,\phi + \omega_z\big) + \dfrac{\gamma_{M_z}J_{zz}\,\mathrm{c}\,\theta}{\mathrm{c}\,\phi}\ \mathrm{sat}(2s_{M_z}, -1, 1)}{J_{zz}}\right)\frac{\mathrm{c}\,\phi}{\mathrm{c}\,\theta}.
\end{aligned}
\tag{16}
$$

Equation 16 can be simplified to:

$$
\begin{bmatrix} \dot{s}_T \\ \dot{s}_{M_x} \\ \dot{s}_{M_y} \\ \dot{s}_{M_z} \end{bmatrix} = -\begin{bmatrix} \gamma_T\ \mathrm{sat}(2s_T, -1, 1) \\ \gamma_{M_x}\ \mathrm{sat}(2s_{M_x}, -1, 1) \\ \gamma_{M_y}\ \mathrm{sat}(2s_{M_y}, -1, 1) \\ \gamma_{M_z}\ \mathrm{sat}(2s_{M_z}, -1, 1) \end{bmatrix}. \tag{17}
$$

Choosing $\gamma_\alpha > 0$, the Lyapunov derivative becomes

$$
\dot{V}_\alpha(x) = -\gamma_\alpha\, s_\alpha\ \mathrm{sat}(2s_\alpha, -1, 1) < 0 \quad \forall\, s_\alpha \neq 0. \tag{18}
$$

So, $\dot{V}_\alpha(x)$ is negative definite, $V_\alpha(x)$ is positive definite, and $V_\alpha(0) = 0$, and the Lyapunov stability requirements are satisfied for V_α in Equation 14. The surfaces s_α thus decay to zero over time, which, from Equation 7, is equivalent to the tracking error decaying to zero. Additionally, within the boundary layer ($|s_\alpha| < 1/2$), $\mathrm{sat}(2s_\alpha, -1, 1) = 2s_\alpha$, and so

$$
\begin{aligned}
\dot{V}_\alpha(x) &= -2\gamma_\alpha\, s_\alpha^2 = -4\gamma_\alpha \left(\tfrac{1}{2}\, s_\alpha^2\right) \\
&= -\rho\, V_\alpha(x), \quad \rho = 4\gamma_\alpha > 0.
\end{aligned}
\tag{19}
$$

■

Outside the boundary layer, $\mathrm{sat}(2s_\alpha, -1, 1) = \mathrm{sgn}(s_\alpha)$, and $\dot{V}_\alpha \leq -\gamma_\alpha |s_\alpha| < 0$, driving the system into the boundary layer in finite time. Combining the two regimes shows that the tracking error is asymptotically stable, with an exponential rate of convergence to the sliding surface.

D. Discrete Sliding Mode Controller

The continuous dynamics assumption is strong. Flight computers and motor speed controllers are both discrete, and have varying duty cycles. We impose an arbitrary 50 Hz zero-order holds on the input and output of the flight controller, and parameterize the discretization to be adjustable depending on the actual discretization of a given system. This violates the continuous-dynamics assumption of [16], and we must derive the dSMC to recover Lyapunov stability in the presence of a discretized plant. We begin by laying out the discrete linear approximation of the nonlinear equations of motion, as presented in [7], where $K_1, \ldots, K_6$ are per-axis drag coefficients (see Table 4 for all coefficients) and $\Omega_r = \omega_1 - \omega_2 + \omega_3 - \omega_4$ is the net rotor speed differential.

$$
x_{k+2} = 2x_{k+1} - x_k - \Delta t\, K_1\, (x_{k+1} - x_k)\, / m
$$

$$+ \Delta t^2 \left(\mathrm{c}\,\phi_k\ \mathrm{s}\,\theta_k\ \mathrm{c}\,\psi_k + \mathrm{s}\,\phi_k\ \mathrm{s}\,\psi_k\right) u_{1,k}/m \tag{20}$$

$$\begin{aligned} y_{k+2} &= 2y_{k+1} - y_k - \Delta t\, K_2\, (y_{k+1} - y_k)\,/m \\ &\quad + \Delta t^2 \left(\mathrm{c}\,\phi_k\ \mathrm{s}\,\theta_k\ \mathrm{s}\,\psi_k - \mathrm{s}\,\phi_k\ \mathrm{c}\,\psi_k\right) u_{1,k}/m \end{aligned} \tag{21}$$

$$\begin{aligned} z_{k+2} &= 2z_{k+1} - z_k - \Delta t\, K_3\, (z_{k+1} - z_k)\,/m \\ &\quad + \Delta t^2 \left(\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k\, u_{1,k}/m - g\right) \end{aligned} \tag{22}$$

$$\begin{aligned} \phi_{k+2} &= 2\phi_{k+1} - \phi_k + (\theta_{k+1} - \theta_k)\,(\psi_{k+1} - \psi_k)\,\frac{(J_{yy} - J_{zz})}{J_{xx}} \\ &\quad + \Delta t^2\, u_{2,k} l/J_{xx} + \Delta t\, J_r\, (\theta_{k+1} - \theta_k)\,\Omega_r/J_{xx} - \Delta t\, K_4 l\, (\phi_{k+1} - \phi_k)\,/J_{xx} \end{aligned} \tag{23}$$

$$\begin{aligned} \theta_{k+2} &= 2\theta_{k+1} - \theta_k + (\psi_{k+1} - \psi_k)\,(\phi_{k+1} - \phi_k)\,\frac{(J_{zz} - J_{xx})}{J_{yy}} \\ &\quad + \Delta t^2\, u_{3,k} l/J_{yy} - \Delta t\, J_r\, (\phi_{k+1} - \phi_k)\,\Omega_r/J_{yy} - \Delta t\, K_5 l\, (\theta_{k+1} - \theta_k)\,/J_{yy} \end{aligned} \tag{24}$$

$$\begin{aligned} \psi_{k+2} &= 2\psi_{k+1} - \psi_k + (\phi_{k+1} - \phi_k)\,(\theta_{k+1} - \theta_k)\,\frac{(J_{xx} - J_{yy})}{J_{zz}} \\ &\quad + \Delta t^2\, u_{4,k}/J_{zz} - \Delta t\, K_6\, (\psi_{k+1} - \psi_k)\,/J_{zz} \end{aligned} \tag{25}$$

Continuing with the problem setup from [7], we now define the sliding surface of the fully actuated subsystem (z and ψ).

$$s_{z,k} = \alpha_z\,(z_k^d - z_k) + (\dot{z}_k^d - \dot{z}_k), \quad s_{\psi,k} = \alpha_\psi\,(\psi_k^d - \psi_k) + (\dot{\psi}_k^d - \dot{\psi}_k). \tag{26}$$

The coefficients α_z and α_ψ are positive constants; and we assume the desired point to be constant, iteration-to-iteration, and so $\dot{z}_k^d = 0$ and $\dot{\psi}_k^d = 0$; where we define $\dot{z}_k = (z_{k+1} - z_k)/\Delta t$, $\dot{\psi}_k = (\psi_{k+1} - \psi_k)\,/\Delta t$ as first-order approximations of the vehicle velocity. This assumption is strong, but it holds for all but finitely many time steps under waypoint commands. To maintain an exponential rate of convergence, we design control laws that satisfy the exponential reaching conditions.

$$s_{z,k+1} - s_{z,k} = \left(-\eta_z s_{z,k}\right)\Delta t, \quad s_{\psi,k+1} - s_{\psi,k} = \left(-\eta_\psi s_{\psi,k}\right)\Delta t \tag{27}$$

where $\eta_z,\ \eta_\psi > 0$. We now show that if $\Delta_{k+1,k} s_i = \left(-\eta_i\, s_{i,k}\right)\Delta t$, the Lyapunov stability requirements presented in [7] are satisfied.

$$\begin{aligned} V_{i,k} &= \tfrac{1}{2} s_{i,k}^2 > 0 \quad \text{(Lyapunov positivity, holds trivially)} \\ \Delta V_{i,k} &\triangleq V_{i,k+1} - V_{i,k} = \tfrac{1}{2}\left(s_{i,k+1}^2 - s_{i,k}^2\right) < 0 \quad \text{(discrete negative definiteness)} \\ &= \tfrac{1}{2}\,(s_{i,k+1} - s_{i,k})\,(s_{i,k+1} + s_{i,k}) \\ &= -\tfrac{1}{2}\,\eta_i\,\Delta t\,|s_{i,k}|\,(2 - \eta_i\,\Delta t)\,|s_{i,k}| < 0 \quad \text{(for appropriate } \eta_i\text{)} \end{aligned} \tag{28}$$

Therefore, if our control laws satisfy the exponential reaching law, the discrete Lyapunov stability requirements hold, and our system exponentially converges to the desired point. This matters for the ballistic-deployment test case, which subjects the controller to large initial deviations.

From here, our work diverges significantly from [7]. We derive different control laws for z and ψ, as the originally proposed control laws do not satisfy the reaching conditions stated above. We assume that the quadrotor is symmetric, and so $J_{xx} = J_{yy}$. This is a standard simplifying assumption and holds for almost all commercial quadrotors.

$$\begin{aligned} u_{1,k} &= \frac{m}{\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}\left(-\alpha_z\,\dot{z}_k + \tfrac{K_3}{m}\,\dot{z}_k + g + \eta_z\, s_{z,k}\right), \\ \Delta_{k+1,k} s_z &= \left(-\eta_z\, s_{z,k}\right)\Delta t, \\ &= \alpha_z\left(z_k^d - z_{k+1}\right) - \tfrac{1}{\Delta t}\Big[2z_{k+1} - z_k + \Delta t^2\left(\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k\,\tfrac{u_{1,k}}{m} - g\right) \\ &\qquad - \tfrac{\Delta t\, K_3}{m}\left(z_{k+1} - z_k\right) - z_{k+1}\Big] - \alpha_z\left(z_k^d - z_k\right) - \tfrac{z_{k+1} - z_k}{\Delta t}, \\ &= \alpha_z\left(z_k - z_{k+1}\right) - \tfrac{1}{\Delta t}\left[\Delta t^2\left(\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k\,\tfrac{u_{1,k}}{m} - g\right) - \tfrac{\Delta t\, K_3}{m}\left(z_{k+1} - z_k\right)\right], \\ &= \alpha_z\left(z_k - z_{k+1}\right) - \Delta t\left(-\alpha_z\,\dot{z}_k + \tfrac{K_3}{m}\,\dot{z}_k + \eta_z\, s_{z,k}\right) + \tfrac{K_3}{m}\left(z_{k+1} - z_k\right), \\ &\ \ \vdots \\ \Delta_{k+1,k} s_z &= \left(-\eta_z\, s_{z,k}\right)\Delta t. \end{aligned} \tag{29}$$

$$
\begin{aligned}
u_{4,k} &= J_{zz}\Big(-\alpha_\psi\,\dot\psi_k + \tfrac{K_6}{J_{zz}}\,\dot\psi_k + \eta_\psi\, s_{\psi,k}\Big),\\
&\vdots\\
s_{\psi,\,k+1} &= \alpha_\psi\big(\psi_k^d - \psi_{k+1}\big) - \Big(\dot\psi_k + \Delta t\big(-\alpha_\psi\,\dot\psi_k + \tfrac{K_6}{J_{zz}}\,\dot\psi_k + \eta_\psi\, s_{\psi,k}\big) - \tfrac{\Delta t\, K_6}{J_{zz}}\,\dot\psi_k\Big),\\
&\vdots\\
\Delta_{k+1,k} s_\psi &= -\,\Delta t\,\eta_\psi\, s_{\psi,k}.
\end{aligned}
\tag{30}
$$

In implementation, the fully-actuated sliding variables are saturated to $s_{z,k},\ s_{\psi,k} \in [-2,\,2]$ before entering the control laws (Table 4). Outside that band the reaching step becomes the constant-rate $\Delta_{k+1,k} s_i = -2\,\eta_i\,\Delta t\ \mathrm{sgn}(s_{i,k})$, which still satisfies the discrete existence condition $|s_{i,k+1}| < |s_{i,k}|$ for $\eta_i\,\Delta t < 2$; the exponential reaching law above is recovered once $|s_{i,k}| \le 2$. In the saturated controller of Section VI.C, the clip is applied to s before the auxiliary-state correction that forms $\tilde{s}$.

Contrary to this formulation, [7] proposed the following control laws for the fully-actuated dimensions of the quadrotor:

$$
u_{1,k} = \frac{m}{\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}\left\{\frac{-\alpha_z \dot z_k + g + K_3 \dot z_k}{m + \eta_z\, s_{z,k}}\right\}
\tag{31}
$$

$$
u_{4,k} = J_{zz}\left\{\frac{-\alpha_\psi \dot\psi_k + K_6 \dot\psi_k}{J_{zz} + \eta_\psi\, s_{\psi,k}}\right\}
\tag{32}
$$

We substitute the two-step extrapolated dynamics along with the proposed control laws above into $s_{i,k+1} - s_{i,k}$ and simplify.

$$
\begin{aligned}
\Delta_{k+1,k} s_z &= -\alpha_z\,\Delta t\,\dot z_k - \Delta t\left[\frac{-\alpha_z\,\dot z_k + g + K_3\,\dot z_k}{m + \eta_z\, s_{z,k}} - g\right] + \tfrac{K_3}{m}\,\Delta t\,\dot z_k\\
&= -\alpha_z\,\Delta t\,\dot z_k + \Delta t\,\frac{\alpha_z\,\dot z_k}{m + \eta_z\, s_{z,k}} - \Delta t\,\frac{g}{m + \eta_z\, s_{z,k}}\\
&\qquad - \Delta t\,\frac{K_3\,\dot z_k}{m + \eta_z\, s_{z,k}} + \frac{\Delta t\, K_3\,\dot z_k}{m} + g\,\Delta t\\
&\neq \big(-\eta_z\, s_{z,k}\big)\,\Delta t
\end{aligned}
\tag{33}
$$

$$
\begin{aligned}
\Delta_{k+1,k} s_\psi &= -\alpha_\psi\,\Delta t\,\dot\psi_k - \Delta t\ \left[\frac{-\alpha_\psi\,\dot\psi_k + K_6\,\dot\psi_k}{J_{zz} + \eta_\psi\, s_{\psi,k}} - \frac{K_6}{J_{zz}}\,\dot\psi_k\right]\\
&= \alpha_\psi\,\Delta t\,\dot\psi_k\left(\frac{1}{J_{zz} + \eta_\psi\, s_{\psi,k}} - 1\right) + \Delta t\, K_6\,\dot\psi_k\left(\frac{1}{J_{zz}} - \frac{1}{J_{zz} + \eta_\psi\, s_{\psi,k}}\right)\\
&\neq \big(-\eta_\psi\, s_{\psi,k}\big)\,\Delta t
\end{aligned}
\tag{34}
$$

Therefore, the control laws proposed in [7] violate the exponential reaching condition, and we proceed using the control laws of Equations 29 and 30. We now derive the control laws for pitch and roll. We enforce the same exponential reaching laws, which, as we proved in Equation 28, satisfy the conditions for Lyapunov stability. For brevity, we derive u_2, the roll control law, which has the same derivation form as u_3.

$$
\begin{aligned}
s_{\phi,k} &= -a_1\,\dot{y}_k + a_2\Big(y_k^d - y_k\Big) - a_3\,\dot{\phi}_k + a_4\Big(\phi_k^d - \phi_k\Big)\\
T_{y,k} &\triangleq \mathrm{c}\,\phi_k\ \mathrm{s}\,\theta_k\ \mathrm{s}\,\psi_k - \mathrm{s}\,\phi_k\ \mathrm{c}\,\psi_k, \quad T_{x,k} \triangleq \mathrm{c}\,\phi_k\ \mathrm{s}\,\theta_k\ \mathrm{c}\,\psi_k + \mathrm{s}\,\phi_k\ \mathrm{s}\,\psi_k\\
g_{1,k} &\triangleq \frac{T_{y,k}}{m}, \quad g_{2,k} \triangleq \frac{T_{x,k}}{m}, \quad \Omega_r \triangleq \omega_1 - \omega_2 + \omega_3 - \omega_4\\
f_{1,k} &\triangleq \frac{1}{J_{xx}}\Big(\dot{\theta}_k\,\dot{\psi}_k\big(J_{yy} - J_{zz}\big) + J_r\,\dot{\theta}_k\,\Omega_r - K_4\,l\,\dot{\phi}_k\Big)\\
f_{2,k} &\triangleq \frac{1}{J_{yy}}\Big(\dot{\psi}_k\,\dot{\phi}_k\big(J_{zz} - J_{xx}\big) - J_r\,\dot{\phi}_k\,\Omega_r - K_5\,l\,\dot{\theta}_k\Big)
\end{aligned} \tag{35}
$$

$$
\begin{aligned}
y_{k+2} - y_{k+1} &= \Delta t\,\dot{y}_k\Big(1 - \tfrac{\Delta t\,K_2}{m}\Big) + \tfrac{\Delta t^2 T_{y,k}}{m}\,u_{1,k} \implies \dot{y}_{k+1} - \dot{y}_k = \Delta t\Big[g_{1,k}\,u_{1,k} - \tfrac{K_2}{m}\,\dot{y}_k\Big]\\
\phi_{k+2} - \phi_{k+1} &= \Delta t\,\dot{\phi}_k + \Delta t^2\Big[f_{1,k} + \tfrac{u_{2,k}\,l}{J_{xx}}\Big] \implies \dot{\phi}_{k+1} - \dot{\phi}_k = \Delta t\Big[f_{1,k} + \tfrac{u_{2,k}\,l}{J_{xx}}\Big]\\
x_{k+2} - x_{k+1} &= \Delta t\,\dot{x}_k\Big(1 - \tfrac{\Delta t\,K_1}{m}\Big) + \tfrac{\Delta t^2 T_{x,k}}{m}\,u_{1,k} \implies \dot{x}_{k+1} - \dot{x}_k = \Delta t\Big[g_{2,k}\,u_{1,k} - \tfrac{K_1}{m}\,\dot{x}_k\Big]\\
\theta_{k+2} - \theta_{k+1} &= \Delta t\,\dot{\theta}_k + \Delta t^2\Big[f_{2,k} + \tfrac{u_{3,k}\,l}{J_{yy}}\Big] \implies \dot{\theta}_{k+1} - \dot{\theta}_k = \Delta t\Big[f_{2,k} + \tfrac{u_{3,k}\,l}{J_{yy}}\Big]
\end{aligned} \tag{36}
$$

$$
\delta y_k \triangleq g_{1,k}\,u_{1,k} - \frac{K_2}{m}\,\dot{y}_k, \quad \delta x_k \triangleq g_{2,k}\,u_{1,k} - \frac{K_1}{m}\,\dot{x}_k \tag{37}
$$

Roll-axis surface:

$$
\begin{aligned}
\Delta_{k+1,k}s_\phi &= \Delta t\left[-a_1\,\delta y_k - a_2\,\dot{y}_k - a_3\left(f_{1,k} + \frac{u_{2,k}\,l}{J_{xx}}\right) - a_4\,\dot{\phi}_k\right] = -\eta_\phi\,s_{\phi,k}\,\Delta t,\\
u_{2,k} &= \frac{J_{xx}}{l\,a_3}\left[-a_1\,\delta y_k - a_2\,\dot{y}_k - a_3\,f_{1,k} - a_4\,\dot{\phi}_k + \eta_\phi\,s_{\phi,k}\right].
\end{aligned} \tag{38}
$$

Pitch-axis surface:

$$
\begin{aligned}
\Delta_{k+1,k}s_\theta &= \Delta t\left[-a_5\,\delta x_k - a_6\,\dot{x}_k - a_7\left(f_{2,k} + \frac{u_{3,k}\,l}{J_{yy}}\right) - a_8\,\dot{\theta}_k\right] = -\eta_\theta\,s_{\theta,k}\,\Delta t,\\
u_{3,k} &= \frac{J_{yy}}{l\,a_7}\left[-a_5\,\delta x_k - a_6\,\dot{x}_k - a_7\,f_{2,k} - a_8\,\dot{\theta}_k + \eta_\theta\,s_{\theta,k}\right].
\end{aligned} \tag{39}
$$

IV. Simulation Design

A. Ballistic Trajectory

We simulate a six degrees of freedom (6-DOF) ballistic trajectory using the equations outlined in Chapter 9 of [17]. For the moment, we use the standard model of a 120mm mortar shell given in [17], due to readily available aerodynamic characteristics and our focus on the family of initial conditions generated by the ballistic trajectory, rather than the particular trajectory. The specific trajectory parameters are given in Table 6. The launch elevation is chosen to place the deployment envelope within the controller's recoverable regime: a shallow (45°) launch produces a fast (≈70 m/s) apogee deploy that exceeds the arrestable envelope, whereas the steeper 64° launch trades range for a gentler (≈45 m/s) deploy.

Table 4 Gains and scalar parameters.

Parameter	Value	Parameter	Value
K_1	0.0001	K_2	0.0001
K_3	0.0001	K_4	0.0012
K_5	0.0012	K_6	0.0012
η_z	7	η_ψ	7
η_ϕ	14	η_θ	14
α_z	6	α_ψ	1
b	5	d	2
Ω^2_{min}	0	Ω^2_{max}	2
s_z, s_ψ clip	± 2	$k_{\xi,\alpha}$	$\eta_\alpha \Delta t$

Table 5 Coefficient definitions a_i.

a_i	Definition	a_i	Definition
a_1	$\dfrac{6\,m}{u_T\ \mathrm{c}\,\psi}$	a_2	$\dfrac{2\,m}{u_T\ \mathrm{c}\,\psi}$
a_3	2	a_4	8
a_5	$-\dfrac{6\,m}{u_T\ \mathrm{c}\,\phi\ \mathrm{c}\,\psi}$	a_6	$-\dfrac{2\,m}{u_T\ \mathrm{c}\,\phi\ \mathrm{c}\,\psi}$
a_7	2	a_8	8

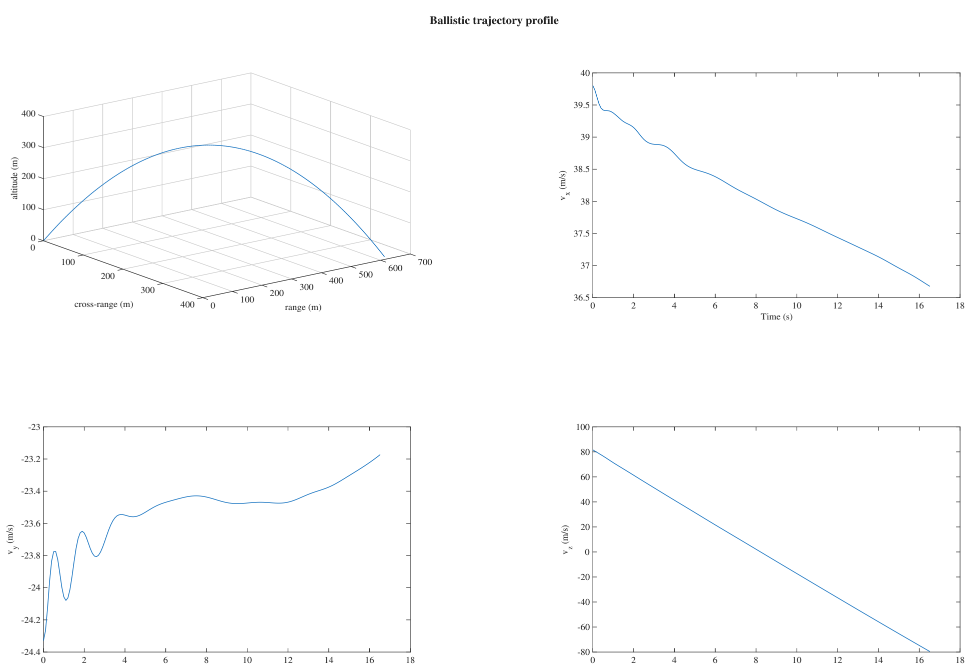

Fig. 2 Mortar shell ballistic trajectory. States sampled along this trajectory are the initial conditions for the drone deployment simulations.

Table 6 Launch parameters for the 120 mm mortar ballistic trajectory used as the deployment-point envelope for the dSMC reachability sweep.

Symbol	Description	Value	Unit
Launch kinematics			
V_0	Muzzle velocity	94	m/s
θ_e	Launch elevation angle	64	deg
θ_a	Launch azimuth angle	15	deg
Initial angular state			
p_0	Initial axial (roll) spin rate	-8.379	rad/s
$\omega_{y,0}$	Initial transverse angular rate (yaw axis)	0.5	rad/s
$\omega_{z,0}$	Initial transverse angular rate (pitch axis)	1	rad/s
α_0	Initial angle of attack (pitch misalignment)	2	deg
β_0	Initial sideslip angle (yaw misalignment)	-0.5	deg
Initial position (range, altitude, cross-range)			
x_0	Range-axis launch position	0	m
y_0	Altitude-axis launch position	0	m
z_0	Cross-range-axis launch position	0	m
Integrator			
$t_{\max}$	Maximum propagation time	300	s

B. Parameters and Configuration

We simulate our system in MATLAB Simulink R2025a. The system is discretized using 50 Hz zero-order hold (ZOH) blocks placed before and after the controller plant. The dynamics are continuously integrated, but sampled using ZOHs. The system is integrated using ‘auto’ solver selection and default settings for all integration parameters. The Simulink models and analysis scripts are available at [18].

V. Simulation Results

A. Ballistic Flight Envelope

At evenly spaced points along the ballistic trajectory, we sampled the vehicle state and ran a 30-second stabilization attempt from each sample, recording which initial conditions each controller could recover from. A trial was counted as successful if all three of the following held for every t in the stabilization window:

$$\sqrt{(x_t - x_f)^2 + (y_t - y_f)^2} \leq 10\,\text{m}, \qquad \frac{|v_t|_2}{|v_0|_2} \leq 2, \qquad \frac{|\omega_t|_2}{\left|[\pi\ \pi\ \pi]^\top\right|_2} \leq 2 \ \ (t > 1\,\text{s}). \tag{40}$$

The first condition bounds the horizontal position error to within 10 m of the target. The second prevents the controller from accelerating the vehicle past twice its initial deployment speed $|v_0|_2$. The third caps the body rotational rate at twice $|[\pi\ \pi\ \pi]^\top|_2 = \pi\sqrt{3} \approx 5.44$ rad/s, the magnitude of a rate vector with π rad/s on every body axis. The rotational-rate bound is evaluated only after the first second of flight: the vehicle is released tumbling, so rate excursions inside that window reflect the ballistic-separation transient the controller is still arresting, not the controller itself. Figure 3 shows that no controller recovers every sampled state once physical control constraints are enforced; the dSMC recovers the largest share (the saturated variant, 8 of 17 sampled states), concentrated in the lower-energy ascending deployments. LQR and PID operate too far from their hover linearization, and cSMC’s continuous-time stability argument does not transfer to the sampled-data implementation.

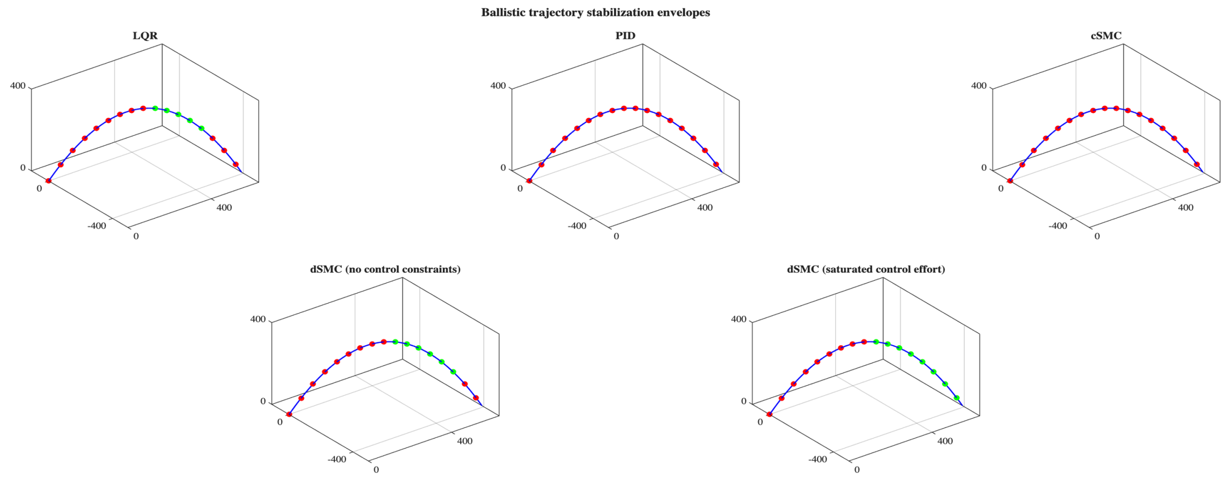

Fig. 3 Ballistic stabilization at test points along the trajectory.

VI. Input Saturation Handling for the Discrete Sliding Mode Controller

The dSMC above produces an unconstrained command $u_k = (T_k, M_{x,k}, M_{y,k}, M_{z,k})^T \in \mathbb{R}^4$. Real motors enforce per-rotor speed limits, whether due to electrical limitations of the motor or its electronic speed controller, or aerostructural limitations of the rotor itself. These limitations define a feasible set $\mathcal{U} \subset \mathbb{R}^4$ for the applied command signals, namely, per-motor rotation speeds. When $u_k \notin \mathcal{U}$, the controller has to project onto $\mathcal{U}$ and account for the resulting tracking deficit. Inside $\mathcal{U}$, and provided no saturation has yet occurred, the closed loop is Lyapunov stable, as the controller coincides with the unconstrained dSMC; once a saturation episode ends, the controller instead converges back to the baseline dSMC as the anti-windup state decays (Section VI.G). Outside $\mathcal{U}$, it relaxes to an ISS bound, which we will show below.

A. Motor Mixing and Saturation Limits

Rotor-speed-squared $\Omega^2 = (\Omega_1^2, \Omega_2^2, \Omega_3^2, \Omega_4^2)^T$ relates to the command via the symmetric-quadrotor mixing matrix

$$u = T_M\,\Omega^2, \qquad T_M = \begin{bmatrix} b & b & b & b \\ 0 & -b & 0 & b \\ -b & 0 & b & 0 \\ d & -d & d & -d \end{bmatrix}, \tag{41}$$

where b is the per-rotor thrust-per-Ω^2 coefficient, d the analogous yaw-torque coefficient, and the roll/pitch moment rows carry an effective moment coefficient that we set numerically equal to b. We work in normalized Ω^2 units with the implementation values $b = 5$, $d = 2$, $\Omega^2_{\text{min}} = 0$, and $\Omega^2_{\text{max}} = 2$ (Table 4); these describe a notional actuator with thrust-to-weight ratio $4b\,\Omega^2_{\text{max}}/(mg) = 5.10$ and hover point $\Omega^2_{\text{hover}} = mg/(4b) = 0.392$ strictly inside the feasible band, and are effective values for the simulated actuator rather than quantities derived from the physical mixing of Equation 1. Extending the component-wise discrete-time saturation framework of Wang et al. [13] to the per-rotor case (which we choose due to the underlying source of input constraints being motor speed limits, and so is more physically connected), $\Omega_i^2 \in [\Omega^2_{\text{min}}, \Omega^2_{\text{max}}]$ defines the feasible set in command space as $\mathcal{U} = T_M \cdot [\Omega^2_{\text{min}}, \Omega^2_{\text{max}}]^4$. Let $\bar{u}_k \in \mathcal{U}$ denote the actually applied (after saturation) command and define the saturation deficit

$$\Delta u_k \triangleq u_k - \bar{u}_k. \tag{42}$$

The rows of T_M are pairwise orthogonal by inspection, so the Gram matrix is diagonal:

$$T_M T_M^T = \operatorname{diag}\Big(4b^2,\, 2b^2,\, 2b^2,\, 4d^2\Big), \quad (T_M T_M^T)^{-1} = \operatorname{diag}\Big(\tfrac{1}{4b^2},\, \tfrac{1}{2b^2},\, \tfrac{1}{2b^2},\, \tfrac{1}{4d^2}\Big). \tag{43}$$

Now, we can scale channel-by-channel in the closed-form projection below. Scaling the command on a single channel scales one column-contribution of $T_M^{-1} u$ in Ω^2-space - one column of T_M^{-1} times the corresponding component of u, the v_α of Equation 46 - and leaves the other three contributions untouched. We adopt the ranking presented in Faessler et al. [11], which ranks roll and pitch above total thrust above yaw; the low priority of yaw is consistent with Mueller and D'Andrea's [12] treatment of yaw rotation as dynamically inessential for trajectory interception. Anecdotally, simulation results agree with this approach. We observed that when the rotor limit passes the priority-allocation feasibility boundary, roll and pitch divergence lead to destabilization and crashes well before yaw or altitude tracking degrades. We therefore adopt the priority order

$$M_x, M_y \succ T \succ M_z, \tag{44}$$

and enforce the control ordering by successively saturating the control channels. As an example, if a control effort can be made feasible by strictly limiting yaw torque, we will use it instead of an alternate strategy which saturates both yaw and collective thrust. Indexing the four control channels as $\alpha \in \{T, M_x, M_y, M_z\}$, the deficit vector Δu_k, auxiliary state ξ_k, and gain matrix $E = \Delta t\, \mathrm{diag}(\eta_T,\ \eta_{M_x},\ \eta_{M_y},\ \eta_{M_z})$ all follow the $\{T, M_x, M_y, M_z\}$ ordering, with the channel gains identified with the dSMC gains of Table 4 through $(\eta_T,\ \eta_{M_x},\ \eta_{M_y},\ \eta_{M_z}) \equiv (\eta_z,\ \eta_\phi,\ \eta_\theta,\ \eta_\psi)$. The numeric bounds of Section VI.F and the residual ratios of Section VI.H both use this identification.

B. Priority-Weighted Allocation

The projection realizing the priority of Equation 44 acts directly on the per-channel contributions in motor space and admits a closed form. Where Faessler et al. [11] achieve the same hierarchy through an iterative mixer with no stability guarantee, the construction here runs in fixed time per tick and admits the Lyapunov stability proof of Section VI.F.

Using $T_M^{-1} = T_M^T (T_M T_M^T)^{-1}$ together with Equation 43,

$$T_M^{-1} = \begin{bmatrix} b & 0 & -b & d \\ b & -b & 0 & -d \\ b & 0 & b & d \\ b & b & 0 & -d \end{bmatrix} \mathrm{diag}\Big(\tfrac{1}{4b^2},\ \tfrac{1}{2b^2},\ \tfrac{1}{2b^2},\ \tfrac{1}{4d^2}\Big) = \begin{bmatrix} \frac{1}{4b} & 0 & -\frac{1}{2b} & \frac{1}{4d} \\ \frac{1}{4b} & -\frac{1}{2b} & 0 & -\frac{1}{4d} \\ \frac{1}{4b} & 0 & \frac{1}{2b} & \frac{1}{4d} \\ \frac{1}{4b} & \frac{1}{2b} & 0 & -\frac{1}{4d} \end{bmatrix}. \tag{45}$$

Applying T_M^{-1} to a command $u = (T,\ M_x,\ M_y,\ M_z)^T$ and collecting columns,

$$\Omega^2 = T_M^{-1} u = \underbrace{\frac{T}{4b}\mathbf{1}}_{v_T} + \underbrace{\frac{M_x}{2b} e_{M_x}}_{v_{M_x}} + \underbrace{\frac{M_y}{2b} e_{M_y}}_{v_{M_y}} + \underbrace{\frac{M_z}{4d} e_{M_z}}_{v_{M_z}}, \tag{46}$$

where $\mathbf{1} = (1,1,1,1)^T$, $e_{M_x} = (0,-1,0,1)^T$, $e_{M_y} = (-1,0,1,0)^T$, $e_{M_z} = (1,-1,1,-1)^T$ are the sign patterns of the rows of T_M. The vectors $v_\alpha \in \mathbb{R}^4$ are the per-channel contributions to motor-speed-squared, distinct from the deficit components c_α used in Section VI.H. The four basis vectors $\{\mathbf{1},\ e_{M_x},\ e_{M_y},\ e_{M_z}\}$ are mutually orthogonal in $\mathbb{R}^4$ (they are columns of T_M^T, whose Gram matrix is diagonal). Separately, the additive structure of Equation 46 maps each command channel to a single contribution v_α, so replacing the command on channel α by $\lambda\, u_\alpha$ scales only v_α by λ and leaves the other three contributions untouched. Each component of $\Omega^2(\lambda)$ is then an affine function of the single scalar λ, and componentwise feasibility $\Omega_i^2(\lambda) \in [\Omega^2_{\min},\ \Omega^2_{\max}]$ reduces to four scalar interval constraints, solved in closed form by

$$\lambda_{\max}(\text{base, dir}) = \max\left(0,\ \min\left(1,\ \min_{i:\,\text{dir}_i \neq 0} \begin{cases} \frac{\Omega^2_{\max} - \text{base}_i}{\text{dir}_i} & \text{if dir}_i > 0 \\ \frac{\Omega^2_{\min} - \text{base}_i}{\text{dir}_i} & \text{if dir}_i < 0 \end{cases}\right)\right), \tag{47}$$

returning 0 if any component with $\text{dir}_i = 0$ has $\text{base}_i \notin [\Omega^2_{\min},\ \Omega^2_{\max}]$. When the base point is itself feasible, the point returned by Equation 47 is feasible; when it is not, the returned scale carries no such guarantee. The trailing feasibility checks in the cascade below are therefore load-bearing rather than redundant: each stage short of the terminal fallback returns its candidate only after re-checking the full motor-space point.

We apply cascaded saturation by successively saturating channels, in ascending order of importance, until we reach a feasible control effort. So, we accept relatively minor errors (some yaw error, or a minor loss of altitude) in exchange for controlling more important channels (orientation, and thus, thrust vector pointing).

1) The squared rotor speed $\Omega^2 = T_M^{-1} u_k$. If $\Omega^2 \in [\Omega^2_{\min},\ \Omega^2_{\max}]^4$, then our control effort satisfies our control limits, and we simply return $\bar{u}_k = u_k$ and $\Delta u_k = 0$, which is the unmodified dSMC control effort.

2) Otherwise, our control effort is infeasible, and we begin the saturation process. We start by scaling the lowest-priority channel M_z by $\alpha_{M_z} \in [0, 1]$ while leaving $T,\ M_x,\ M_y$ at their commanded values:

$$\Omega^2(\alpha_{M_z}) = \underbrace{\nu_T + \nu_{M_x} + \nu_{M_y}}_{\text{base}} + \alpha_{M_z} \underbrace{\nu_{M_z}}_{\text{dir}}. \tag{48}$$

With $\alpha_{M_z} = \lambda_{\max}(\text{base},\ \nu_{M_z})$, return $\bar{u}_k = (T,\ M_x,\ M_y,\ \alpha_{M_z} M_z)^T$ if feasible. In regime, we begin to sacrifice some yaw control, but retain altitude (collective thrust) and attitude (roll/pitch) control.

3) If yaw scaling is unable to produce a feasible control effort, we temporarily abandon yaw control and begin scaling collective thrust by $\beta_T \in [0, 1]$, which affects our altitude control:

$$\Omega^2(\beta_T) = \underbrace{\nu_{M_x} + \nu_{M_y}}_{\text{base}} + \beta_T \underbrace{\nu_T}_{\text{dir}}. \tag{49}$$

With $\beta_T = \lambda_{\max}(\nu_{M_x} + \nu_{M_y},\ \nu_T)$, return $\bar{u}_k = (\beta_T T,\ M_x,\ M_y,\ 0)^T$ if the control effort is feasible. In this case, we maintain full roll and pitch authority, limited altitude authority, and completely discard our yaw authority.

4) If thrust scaling is infeasible, then zero collective thrust and zero yaw cannot accommodate the demanded (M_x, M_y). In this case, we scale the joint roll/pitch contribution by $\gamma = \lambda_{\max}(0,\ \nu_{M_x} + \nu_{M_y}) \in [0, 1]$ and return $\bar{u}_k = (0,\ \gamma M_x,\ \gamma M_y,\ 0)^T$. This terminal stage assumes $\Omega^2_{\min} = 0$; with any strictly positive idle floor, no point on the zero-thrust ray is feasible and the closed form admits no feasible scale. Even under that assumption, the closed form is blunt: the components of the direction $\nu_{M_x} + \nu_{M_y} = \frac{1}{2b}(-M_y,\ -M_x,\ M_y,\ M_x)^T$ occur in $\pm$ pairs, so at least one is strictly negative whenever $(M_x, M_y) \neq (0, 0)$, and with a zero base Equation 47 returns $\gamma = 0$. The terminal fallback is therefore $\bar{u}_k = 0$, a full motor cutoff that retains no control authority on any channel for that tick, rather than a scaled roll/pitch command. The severity of this stage should intuitively indicate why all stability proofs of our saturated controller require saturation to be transient - upon reaching stage 4 the vehicle coasts unactuated, and recovery is possible only once the commanded moments shrink enough for an earlier stage of the cascade to return a feasible effort.

C. Auxiliary State and Modified Sliding Variable

The projection $u_k \mapsto \bar{u}_k$ generates $\Delta u_k \neq 0$ whenever saturation is required. Were we to simply substitute $u_k \mapsto \bar{u}_k$ into our previous derivation of the sliding surface, the control deficit would violate the reaching condition $\Delta s_k = -\eta_\alpha \Delta t\, s_k$, and invalidate our stability proofs. Therefore, we need to modify the sliding surface using auxiliary states in order to account for this control deficit. We follow the the basic concepts of the discrete-time anti-windup compensator of Wang et al. [13], in which an auxiliary state is driven by the saturation deficit $\sigma(u) - u$ to recover stability under input saturation. Adapting the continuous-time sliding-mode auxiliary-state constructions of [8, 9] into discrete time across all four actuator channels, we introduce, for each $\alpha \in \{T, M_x, M_y, M_z\}$, a scalar state ξ_α with update

$$\xi_{\alpha\, k+1} = (1 - k_{\xi\ \alpha})\, \xi_{\alpha\, k} + \Delta u_{\alpha\, k}, \qquad 0 < k_{\xi\ \alpha} < 2, \tag{50}$$

i.e., a discrete-time first-order filter driven by the per-channel deficit, pole at $1 - k_{\xi\ \alpha}$. In matrix form, $\xi_{k+1} = (I - K_\xi)\, \xi_k + \Delta u_k$ with $K_\xi = \mathrm{diag}(k_{\xi\ \alpha})$. The diagonal structure obtained by using one scalar state per actuator channel (rather than the vector-valued second-order auxiliary subsystem in [9] or the attitude-only per-channel construction in [8]) is what enables the diagonal Lyapunov decomposition in Section VI.F. Defining

$$\tilde{s}_k \ \triangleq\ s_k - D_k\, \xi_k, \tag{51}$$

where $D_k \in \mathbb{R}^{4\times4}$ is the deficit-coupling matrix derived in the following subsection, the baseline dSMC control laws are reused verbatim with $s_\alpha \mapsto \tilde{s}_\alpha$ substituted in the reaching-law terms of Equations 29–30 and their underactuated analogs; the equivalent-control terms are unchanged. So, $D_k\, \xi_k$ is the cumulative effect of past deficits on the sliding manifold, and subtracting it from s_k gives the controller an estimate of what the sliding variable would have been without saturation. The cancellation occurs in Equation 76.

Each control channel of the auxiliary state has its own contraction variable, $1 - k_{\xi\ \alpha}$. We choose

$$k_{\xi\ \alpha} = \eta_\alpha\, \Delta t, \tag{52}$$

which is the same rate at which the unconstrained reaching law drives s to zero. This way, we can reuse the dSMC gains directly via Equation 52, as in the zero-control-deficit case, we recover the unsaturated dSMC controller. Both blocks of the closed-loop matrix $\mathcal{A}_k$ (Equation 75) then decay at consistent per-channel rates, and the steady-state residual analysis of Section VI.H becomes Δt-independent ratios. When the commanded per-motor speeds lie within the feasible range, $\Delta u_k = 0$ forces ξ to contract to zero, and the modified sliding variable reduces to the baseline s_k (Section VI.G).

D. Deficit-Coupling Matrix

We now derive the auxiliary state - control deficit coupling matrix D_k. Two related coefficients enter the analysis and must be distinguished. Define

$$A_{ij} \triangleq \frac{\partial(\Delta s_i)}{\partial u_j} \qquad (\textit{formula coefficient}\text{: how commanded } u_j \text{ enters } \Delta s_i) \tag{53}$$

$$C_{ij} \triangleq \frac{\partial(\Delta s_i)}{\partial(\Delta u_j)} \qquad (\textit{deficit-response coefficient}\text{: how the deficit propagates}). \tag{54}$$

Decompose the intended surface change as

$$\Delta s_i\big|_{\text{commanded}} = A_{ij}\, u_j + (\text{terms independent of } u_j). \tag{55}$$

The saturated input is $\bar{u}_j = u_j - \Delta u_j$, and so the surface change corresponding to this saturated input is

$$\Delta s_i\big|_{\text{actual}} = A_{ij}\, \bar{u}_j + (\text{rest}) = \Delta s_i\big|_{\text{commanded}} - A_{ij}\, \Delta u_j. \tag{56}$$

The $\partial/\partial(\Delta u_j)$ on the right-hand side gives

$$C = -A. \tag{57}$$

The matrix used in the substitution Equation 51 must equal C for the deficit to cancel in the $\tilde{s}$ dynamics (verified in Equation 76 below), so

$$D = C = -A. \tag{58}$$

The four nonzero entries follow from direct differentiation of the discrete sliding-variable increments. For the fully-actuated thrust $\rightarrow s_z$ entry $D_{1,1}$, the two-step extrapolated altitude dynamics of [7] give the velocity increment

$$\dot{z}_{k+1} - \dot{z}_k = \Delta t \left[\frac{\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}{m}\, u_{1,k} \ - \ g \ - \ \frac{K_3}{m}\, \dot{z}_k \right]. \tag{59}$$

With $s_{z,k} = \alpha_z (z_k^d - z_k) + (\dot{z}_k^d - \dot{z}_k)$ and constant reference state inputs (as is the case when landing, where the reference state is simply the landing point at rest), the increment is:

$$\Delta_{k+1,k} s_z = -\alpha_z\, \Delta t\, \dot{z}_k \ - \ \Delta t \left[\tfrac{\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}{m}\, u_{1,k} \ - \ g \ - \ \tfrac{K_3}{m}\, \dot{z}_k \right]. \tag{60}$$

Only the $-\Delta t\ \mathrm{c}\,\phi_k\, \mathrm{c}\,\theta_k\, u_{1,k}/m$ term contains $u_{1,k}$, giving

$$A_{1,1} \ = \ -\frac{\Delta t\ \mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}{m}, \qquad D_{1,1} \ = \ \frac{\Delta t\ \mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}{m}. \tag{61}$$

For the fully-actuated yaw $\rightarrow s_\psi$ entry $D_{4,4}$, the two-step extrapolated yaw dynamics give $\dot{\psi}_{k+1} - \dot{\psi}_k = \Delta t\, [\, u_{4,k}/J_{zz} - (K_6/J_{zz})\, \dot{\psi}_k\,]$. The same procedure applied to $s_{\psi,k} = \alpha_\psi (\psi_k^d - \psi_k) + (\dot{\psi}_k^d - \dot{\psi}_k)$ yields

$$A_{4,4} \ = \ -\frac{\Delta t}{J_{zz}}, \qquad D_{4,4} \ = \ \frac{\Delta t}{J_{zz}}. \tag{62}$$

The underactuated roll entries $D_{2,1}, D_{2,2}$ follow from

$$s_{\phi,k} = -a_1\, \dot{y}_k + a_2 (y_k^d - y_k) - a_3\, \dot{\phi}_k + a_4 (\phi_k^d - \phi_k). \tag{63}$$

With a constant reference state (present in the landing case, with the landing point at rest as the desired state), the surface evolution expands as

$$\Delta_{k+1,k} s_\phi = -a_1\, (\dot{y}_{k+1} - \dot{y}_k) \ - \ a_2\, (y_{k+1} - y_k) - a_3\, (\dot{\phi}_{k+1} - \dot{\phi}_k) \ - \ a_4\, (\phi_{k+1} - \phi_k). \tag{64}$$

Substituting the two-step linearly extrapolated dynamics,

$$\dot{y}_{k+1} - \dot{y}_k = \Delta t\Big[\, g_{1,k}\, u_{1,k} - \tfrac{K_2}{m}\, \dot{y}_k \,\Big] \;=:\; \Delta t\, \delta y_k, \tag{65}$$

$$\dot{\phi}_{k+1} - \dot{\phi}_k = \Delta t\Big[\, f_{1,k} + \tfrac{u_{2,k}\, l}{J_{xx}} \,\Big], \tag{66}$$

$$y_{k+1} - y_k = \Delta t\, \dot{y}_k, \qquad \phi_{k+1} - \phi_k = \Delta t\, \dot{\phi}_k, \tag{67}$$

gives

$$\Delta_{k+1,k} s_\phi = \Delta t\Big[-a_1\, \delta y_k \;-\; a_2\, \dot{y}_k \;-\; a_3\Big(f_{1,k} + \tfrac{u_{2,k}\, l}{J_{xx}}\Big) \;-\; a_4\, \dot{\phi}_k \,\Big]. \tag{68}$$

Two control-dependent terms appear: $\delta y_k = g_{1,k}\, u_{1,k} - K_2\, \dot{y}_k/m$ is dependent on $u_{1,k}$, thrust, and $a_3 u_{2,k} l/J_{xx}$ is dependent on $u_{2,k}$, the roll moment. Differentiating,

$$\Delta_{k+1,k} s_\phi\big|_u \;=\; -\Delta t\, a_1\, g_{1,k}\, u_{1,k} \;-\; \frac{\Delta t\, a_3\, l}{J_{xx}}\, u_{2,k}, \tag{69}$$

yielding

$$A_{2,1} = -\Delta t\, a_1\, g_{1,k}, \qquad A_{2,2} = -\frac{\Delta t\, a_3\, l}{J_{xx}}, \tag{70}$$

and, by $D = -A$, $D_{2,1} = \Delta t\, a_1\, g_{1,k}$, $D_{2,2} = \Delta t\, a_3\, l/J_{xx}$. One implementation point matters here: the δy_k term inside the wrapped roll law (and its pitch analog δx_k) must be evaluated with the commanded $u_{1,k}$, not the saturated $\bar{u}_{1,k}$, since Equation 69 differentiates with respect to the commanded input. Evaluating it with $\bar{u}_{1,k}$ changes the effective $D_{2,1}$ and $D_{3,1}$ and leaves an unmodeled residual in the cancellation of Section VI.E. The pitch entries $D_{3,1}, D_{3,3}$ are structurally the same, but replacing $\{a_1, a_2, a_3, a_4\}$ with $\{a_5, a_6, a_7, a_8\}$ respectively, $g_{1,k}$ with $g_{2,k}$, $\dot{y}_k$ with $\dot{x}_k$, $u_{2,k}$ with $u_{3,k}$, and J_{xx} with J_{yy}.

$$A_{3,1} = -\Delta t\, a_5\, g_{2,k}, \qquad A_{3,3} = -\frac{\Delta t\, a_7\, l}{J_{yy}}, \tag{71}$$

giving $D_{3,1} = \Delta t\, a_5\, g_{2,k}$ and $D_{3,3} = \Delta t\, a_7\, l/J_{yy}$.

The remaining entries of A (and hence D) are zero, as thrust does not appear in the yaw dynamics ($A_{4,1} = 0$), and roll moment neither appears in Δs_θ (orthogonal underactuated channel) nor in $\Delta s_z, \Delta s_\psi$ (fully-actuated channels). The same reasoning eliminates the remaining off-diagonals in columns 2–4. Only column 1 (thrust) and the four diagonal elements are nonzero. Collecting and applying $D = -A$, with rows ordered $\{s_z,\ s_\phi,\ s_\theta,\ s_\psi\}$ and columns $\{u_T, u_{M_x}, u_{M_y}, u_{M_z}\}$,

$$D_k = \Delta t \begin{bmatrix} \dfrac{\mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k}{m} & 0 & 0 & 0 \\ a_1\, g_{1,k} & \dfrac{a_3\, l}{J_{xx}} & 0 & 0 \\ a_5\, g_{2,k} & 0 & \dfrac{a_7\, l}{J_{yy}} & 0 \\ 0 & 0 & 0 & \dfrac{1}{J_{zz}} \end{bmatrix}. \tag{72}$$

Diagonal elements are positive: $D_{2,2}$ and $D_{3,3}$ since $a_3 = a_7 > 0$, $D_{4,4} = \Delta t/J_{zz} > 0$ unconditionally, and $D_{1,1} = \Delta t\ \mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k/m > 0$ provided $|\phi_k|$, $|\theta_k| < 90°$, an attitude condition that we fold into the explicit trajectory envelope of Section VI.F. At hover, $D_{2,1}$ and $D_{3,1} \to 0$, leaving D_k diagonal. Because a_1, a_2, a_5, a_6 depend only on the commanded u_T and the attitude, $D_{1,1}$ and $D_{4,4}$ are computed from state only, and u_T from $\tilde{s}_z = s_z - D_{1,1}\, \xi_1$, then a_1, a_2, a_5, a_6 from u_T, then $D_{2,1}, D_{2,2}, D_{3,1}, D_{3,3}$ to complete the matrix. With $D_{1,1} > 0$, a positive thrust deficit ($\xi_T > 0$, upper-bound saturation on u_T) reduces $\tilde{s}_z = s_z - D_{1,1}\, \xi_T$ below s_z, which in turn reduces commanded u_T. The symmetric statement holds for lower bound saturation and for the other moment channels. Therefore, we verify that the saturation is being applied in the correct direction.

E. Schur Stability of the Augmented System

Deriving now the augmented closed-loop system with $(\tilde{s}, \xi)$, we aim to establish Schur (discrete-time Hurwitz) stability. The exponential reaching laws for attitude and altitude require/enforce $\Delta s_k = -E\, s_k$, and so substituting $s \mapsto \tilde{s}$ in the reaching-law term carries the same identity forward into $\tilde{s}$:

$$\Delta s_k\big|_{\text{commanded}} \;=\; -E\, \tilde{s}_k, \qquad E \triangleq \Delta t\, \mathrm{diag}(\eta_T,\ \eta_{M_x},\ \eta_{M_y},\ \eta_{M_z}). \tag{73}$$

The actually-applied input is $\bar{u}_k = u_k - \Delta u_k$. By Equation 57,

$$\Delta s_k = -E\,\tilde{s}_k + D_k\,\Delta u_k. \tag{74}$$

The control deficit enters into s via D_k. However, it does not enter into $\tilde{s}$, as shown below. Combining Equations 51, 50, and 74 with $X_k \triangleq (\tilde{s}_k^T, \xi_k^T)^T \in \mathbb{R}^8$,

$$X_{k+1} = \mathcal{A}_k\, X_k + \mathcal{B}\,\Delta u_k, \quad \mathcal{A}_k = \begin{bmatrix} I-E & D_k\,K_\xi \\ 0 & I-K_\xi \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0 \\ I \end{bmatrix}. \tag{75}$$

Two features of $\mathcal{A}_k$ drive the analysis: (i) the lower-left zero block, so that the deficit does not appear in the $\tilde{s}$ row, and (ii) the upper-right block $D_k K_\xi$, the residual cross-coupling, which decays at the ξ-rate.

To check the zero block in $\mathcal{A}_k$, we expand $\tilde{s}_{k+1}$ and take $D_{k+1} \approx D_k$:

$$\begin{aligned}
\tilde{s}_{k+1} &= s_{k+1} - D_k\,\xi_{k+1} && \text{(Eq. 51)}\\
&= s_k + \Delta s_k - D_k\,\xi_{k+1} && \text{(unrolling } s_{k+1})\\
&= s_k + (-E\,\tilde{s}_k + D_k\,\Delta u_k) && \\
&\quad - D_k\big[(I-K_\xi)\,\xi_k + \Delta u_k\big] && \text{(Eqs. 74, 50)}\\
&= (s_k - D_k\,\xi_k) - E\,\tilde{s}_k + D_k\,K_\xi\,\xi_k && (D_k\,\Delta u_k \text{ cancels})\\
&= (I-E)\,\tilde{s}_k + D_k\,K_\xi\,\xi_k. && \text{(definition of } \tilde{s}_k)
\end{aligned} \tag{76}$$

The two $D_k\,\Delta u_k$ contributions (one from the saturated plant, one from the auxiliary update) have opposite signs and cancel out. So, the deficit only enters the ξ equation, and Δu_k is decoupled from $\tilde{s}$. This decoupling permits the diagonal Lyapunov decomposition of Section VI.F.

The matrix $\mathcal{A}_k$ is block upper-triangular, so $\sigma(\mathcal{A}_k) = \sigma(I-E) \cup \sigma(I-K_\xi)$, and both blocks are diagonal:

$$\sigma(\mathcal{A}_k) = \{1 - \eta_\alpha\,\Delta t\}_\alpha \;\cup\; \{1 - k_{\xi\,\alpha}\}_\alpha. \tag{77}$$

Schur stability ($|\lambda| < 1$) simply becomes:

$$0 < \eta_\alpha\,\Delta t < 2, \qquad 0 < k_{\xi\,\alpha} < 2, \qquad \forall\,\alpha. \tag{78}$$

Plant inertia parameters $(m, J_{xx}, J_{yy}, J_{zz})$ enter only through D_k, which lives in the off-diagonal block and does not affect the spectrum, so the Schur *condition* is parameter-uniform. The closed-loop decay constants are not: they depend on $M_D \triangleq \sup_k |D_k|$ through the ISS decay rate ν of Section VI.F.

Two steps above are implicit and deserve explicit justification, since for a general linear time-varying (LTV) system, pointwise Schur eigenvalues do not imply stability: the cancellation in Equation 76 takes $D_{k+1} \approx D_k$, and the spectrum in Equation 77 is that of the frozen-time matrix $\mathcal{A}_k$. Both are rescued by the structure of $\mathcal{A}_k$. Because the diagonal blocks are constant and D_k enters only the off-diagonal block, the frozen-time spectrum is k-independent, and the LTV system is exponentially stable whenever $M_D < \infty$: ξ decays autonomously through the constant block $I - K_\xi$, and $\tilde{s}$ obeys a constant-coefficient stable recursion driven by the bounded input $D_k\,K_\xi\,\xi_k$, plus the remainder $(D_k - D_{k+1})\,\xi_{k+1}$ from the $D_{k+1} \approx D_k$ step, which is likewise $O(|\xi_k|)$ and vanishes with ξ. The ISS-Lyapunov inequality of Section VI.F is the formal certificate of this claim; the spectrum discussion above is diagnostic.

F. ISS-Lyapunov Stability

Take the candidate Lyapunov function

$$V_k = \tilde{s}_k^T\,W_s\,\tilde{s}_k + \rho\,\xi_k^T\,W_\xi\,\xi_k, \qquad W_s, W_\xi \succ 0 \text{ diagonal}, \;\rho > 0, \tag{79}$$

and compute $\Delta V_k \triangleq V_{k+1} - V_k$ by substituting the closed-loop state updates from (Equations 76 and 50) into V_{k+1}. The construction follows the discrete-time ISS-Lyapunov framework of Jiang and Wang [14].

1) Substituting $\tilde{s}_{k+1} = (I-E)\,\tilde{s}_k + D_k\,K_\xi\,\xi_k$ from Equation 76 into the first quadratic term of V_{k+1},

$$\begin{aligned}
\tilde{s}_{k+1}^T W_s\,\tilde{s}_{k+1} &= \tilde{s}_k^T (I-E)^T W_s (I-E)\,\tilde{s}_k \\
&\quad + 2\,\tilde{s}_k^T (I-E)^T W_s\,D_k K_\xi\,\xi_k \\
&\quad + \xi_k^T K_\xi^T D_k^T W_s D_k K_\xi\,\xi_k.
\end{aligned} \tag{80}$$

2) Substituting $\xi_{k+1} = (I - K_\xi)\,\xi_k + \Delta u_k$ into the second quadratic term,

$$\begin{aligned}\rho\,\xi_{k+1}^T W_\xi\,\xi_{k+1} = \rho\,\xi_k^T (I - K_\xi)^T W_\xi (I - K_\xi)\,\xi_k \\ + 2\rho\,\xi_k^T (I - K_\xi)^T W_\xi\,\Delta u_k \\ + \rho\,\Delta u_k^T W_\xi\,\Delta u_k.\end{aligned} \tag{81}$$

3) Combining the two expansions and subtracting V_k,

$$\begin{aligned}\Delta V_k = &\underbrace{\tilde{s}_k^T\big((I - E)^T W_s (I - E) - W_s\big)\tilde{s}_k}_{\text{(a) } \tilde{s}\text{-decay}} \\ &+ \underbrace{2\,\tilde{s}_k^T (I - E)^T W_s\,D_k\,K_\xi\,\xi_k}_{\text{(b) } \tilde{s}\text{–}\xi \text{ cross}} \\ &+ \underbrace{\xi_k^T K_\xi^T D_k^T W_s D_k K_\xi\,\xi_k}_{\text{(c) cross-square residual}} \\ &+ \underbrace{\rho\,\xi_k^T\big((I - K_\xi)^T W_\xi (I - K_\xi) - W_\xi\big)\xi_k}_{\text{(d) } \xi\text{-decay}} \\ &+ \underbrace{2\rho\,\xi_k^T (I - K_\xi)^T W_\xi\,\Delta u_k}_{\text{(e) } \xi\text{–}\Delta u \text{ cross}} \\ &+ \underbrace{\rho\,\Delta u_k^T W_\xi\,\Delta u_k}_{\text{(f) deficit penalty}}.\end{aligned} \tag{82}$$

4) Using $1 - (1 - x)^2 = x(2 - x)$, the decay terms (a) and (d) become

$$(I - E)^T W_s (I - E) - W_s = -\mathrm{diag}\Big(\eta_\alpha \Delta t\,(2 - \eta_\alpha \Delta t)\,(W_s)_{\alpha\alpha}\Big) \prec 0, \tag{83}$$

negative definite under Equation 78; the decay rate is maximized at $\eta_\alpha \Delta t = 1$ and vanishes at $\eta_\alpha \Delta t \in \{0, 2\}$. We can also write,

$$(I - K_\xi)^T W_\xi (I - K_\xi) - W_\xi = -\mathrm{diag}\Big(k_{\xi\,\alpha}\,(2 - k_{\xi\,\alpha})\,(W_\xi)_{\alpha\alpha}\Big) \prec 0. \tag{84}$$

5) Apply Young's inequality to absorb the $\tilde{s}$–ξ cross-term (b). For any $\mu > 0$ and positive-definite P,

$$2\,u^T P\,v \;\le\; \mu\,u^T P\,u \;+\; \tfrac{1}{\mu}\,v^T P\,v. \tag{85}$$

Apply with $u = (I - E)\tilde{s}_k$, $P = W_s$, $v = D_k K_\xi \xi_k$,

$$\text{(b)} \;\le\; \mu\,\tilde{s}_k^T (I - E)^T W_s (I - E)\,\tilde{s}_k \;+\; \tfrac{1}{\mu}\,\xi_k^T K_\xi^T D_k^T W_s D_k K_\xi\,\xi_k. \tag{86}$$

The first Young term carries the quadratic form $(I - E)^T W_s (I - E)$, *not* the decay term (a), so its per-channel coefficient is $+\mu\,(1 - \eta_\alpha \Delta t)^2\,(W_s)_{\alpha\alpha}$. Combining it with (a) gives the net diagonal coefficient $-\big[\eta_\alpha \Delta t\,(2 - \eta_\alpha \Delta t) - \mu\,(1 - \eta_\alpha \Delta t)^2\big](W_s)_{\alpha\alpha}$ on each $\tilde{s}_\alpha^2$, which is negative definite iff the bracket is positive on every channel:

$$0 < \mu < \min_\alpha \frac{\eta_\alpha \Delta t\,(2 - \eta_\alpha \Delta t)}{(1 - \eta_\alpha \Delta t)^2}. \tag{87}$$

Hence $\mu \in (0, 1)$ alone is insufficient: the absorbed form scales as $(1 - \eta_\alpha \Delta t)^2$, not $\eta_\alpha \Delta t\,(2 - \eta_\alpha \Delta t)$. For the gains of Table 4 at 50 Hz ($\eta_z \Delta t = \eta_\psi \Delta t = 0.14$, $\eta_\phi \Delta t = \eta_\theta \Delta t = 0.28$), the binding (z, ψ) channels require $\mu < 0.35$. The second Young term adds to the cross-square residual (c).

6) Apply Young's inequality once more, now to absorb the ξ–Δu cross-term (e), with $u = (I - K_\xi)\xi_k$, $v = \Delta u_k$, $P = \rho W_\xi$, parameter $\nu_2 > 0$,

$$\text{(e)} \;\le\; \rho\nu_2\,\xi_k^T (I - K_\xi)^T W_\xi (I - K_\xi)\,\xi_k \;+\; \tfrac{\rho}{\nu_2}\,\Delta u_k^T W_\xi\,\Delta u_k. \tag{88}$$

The first piece carries the form $(I - K_\xi)^T W_\xi (I - K_\xi)$, contributing $+\rho\nu_2\,(1 - k_{\xi\,\alpha})^2 (W_\xi)_{\alpha\alpha}$ per channel; as with μ in Equation 87, this erodes (rather than cleanly rescales by $1 - \nu_2$) the ξ-decay margin (d), and the ξ-channel stays negative definite only for

$$0 < \nu_2 < \min_\alpha \frac{k_{\xi\,\alpha}\,(2 - k_{\xi\,\alpha})}{(1 - k_{\xi\,\alpha})^2}. \tag{89}$$

Under the matched gains of Equation 52, $k_{\xi\,\alpha} = \eta_\alpha \Delta t$ and this bound coincides with Equation 87: both equal 0.35 for the Table 4 gains at 50 Hz, binding again on the (z, ψ) channels. The second piece adds to the deficit penalty (f). We fix μ and ν_2 within Equations 87 and 89 *before* selecting ρ in the next step, because the domination condition there must hold against the ξ-decay margin that survives this erosion.

7) With μ and ν_2 fixed, and provided $M_D = \sup_k |D_k| < \infty$ (assumption (iii) below), the sum of the cross-square residual (c) and the leftover from the first Young absorption (Equation 86) is bounded by $\left[(1 + 1/\mu)\, M_D^2\, |K_\xi|^2\, |W_s|\right] |\xi_k|^2$. We then choose ρ large enough that the *eroded* ξ-decay dominates this bound,

$$\rho \min_\alpha \left[k_{\xi\,\alpha}\,(2 - k_{\xi\,\alpha}) - \nu_2\,(1 - k_{\xi\,\alpha})^2\right] (W_\xi)_{\alpha\alpha} \;>\; (1 + \tfrac{1}{\mu})\, M_D^2\, |K_\xi|^2\, |W_s|, \tag{90}$$

which is achievable for any finite M_D, since Equation 89 keeps the bracket positive on every channel. The combined contribution of terms (a)–(e), after both absorptions, is then negative definite in $(\tilde{s}, \xi)$. Collecting,

$$\Delta V_k \;\le\; -\nu\, V_k \;+\; \gamma\, |\Delta u_k|^2, \tag{91}$$

where

$$\nu = \frac{1}{\max\!\left(|W_s|,\; \rho\, |W_\xi|\right)}\, \min\!\Big(\min_\alpha \left[\eta_\alpha \Delta t\, (2 - \eta_\alpha \Delta t) - \mu\, (1 - \eta_\alpha \Delta t)^2\right] (W_s)_{\alpha\alpha},$$
$$\min_\alpha \rho \left[k_{\xi\,\alpha}\,(2 - k_{\xi\,\alpha}) - \nu_2\,(1 - k_{\xi\,\alpha})^2\right] (W_\xi)_{\alpha\alpha} \;-\; \left(1 + \tfrac{1}{\mu}\right) M_D^2\, |K_\xi|^2\, |W_s| \Big), \tag{92}$$

$$\gamma = \rho\,\left(1 + \tfrac{1}{\nu_2}\right) |W_\xi|. \tag{93}$$

These constants are exact rather than order-of-magnitude: the first branch of the minimum is the per-channel $\tilde{s}$-decay that survives the first Young absorption, the second is the eroded ξ-decay minus the absorbed cross-square bound, and so $\nu > 0$ holds precisely when Equations 87, 89, and 90 do. Equation 91 is the discrete-time ISS-Lyapunov inequality in the standard form (cf. Jiang and Wang [14], Definition 3.2 and Example 3.4) used throughout the saturated-systems literature.

Under (i) the Schur condition (Equation 78), (ii) hover-trim feasibility $u_{\text{hover}} \in \operatorname{int}(\mathcal{U})$, and (iii) $M_D = \sup_k |D_k| < \infty$, the closed-loop system satisfies:

1) If $\Delta u_k = 0$ for all k, then $V_k \le (1 - \nu)^k V_0$ and $(\tilde{s}_k, \xi_k) \to 0$ exponentially.
2) If $\Delta u_k \to 0$, then $V_k \to 0$ and $(\tilde{s}_k, \xi_k) \to 0$ asymptotically; the controller returns to optimal dSMC as the control deficit decreases.
3) If $\sup_k |\Delta u_k|^2 \le D_{\text{sup}}$, then $\limsup_k V_k \le \gamma D_{\text{sup}}/\nu$, i.e., the controller (and the system response) is ultimately bounded.

Assumptions (i) and (ii) are directly checkable from the gains and the feasible set. Assumption (iii) is not: it is nontrivial precisely in the tumbling regime this paper targets, and we state it as an explicit trajectory envelope rather than leaving it implicit. From Table 5,

$$D_{2,1} = \Delta t\, a_1\, g_{1,k} = \frac{6\,\Delta t\, T_{y,k}}{u_T\ \mathrm{c}\,\psi_k}, \qquad D_{3,1} = \Delta t\, a_5\, g_{2,k} = -\frac{6\,\Delta t\, T_{x,k}}{u_T\ \mathrm{c}\,\phi_k\ \mathrm{c}\,\psi_k}, \tag{94}$$

both of which grow without bound as the commanded $u_T \to 0$ or as $\mathrm{c}\,\psi_k \to 0$ (respectively $\mathrm{c}\,\phi_k\ \mathrm{c}\,\psi_k \to 0$), and a ballistically launched, tumbling vehicle can pass near both. We therefore require: there exist $\epsilon_u, \epsilon_a > 0$ such that, along closed-loop trajectories,

$$|u_T| \ge \epsilon_u, \qquad |\mathrm{c}\,\psi_k| \ge \epsilon_a, \qquad |\mathrm{c}\,\phi_k\ \mathrm{c}\,\psi_k| \ge \epsilon_a, \qquad |\phi_k|, |\theta_k| < 90^\circ, \tag{95}$$

where the first three conditions bound the cross-coupling magnitudes and the last keeps $D_{1,1} > 0$, as used by the deficit-direction argument of Section VI.D. Under Equation 95, every nonzero entry of D_k is bounded - $|T_{x,k}|,\ |T_{y,k}| \le 1$, since each is the inner product of $(\mathrm{c}\,\phi_k\ \mathrm{s}\,\theta_k,\ \mathrm{s}\,\phi_k)$, of norm at most one, with a unit vector - and bounding the spectral norm by the Frobenius norm over the six nonzero entries gives the explicit certificate

$$M_D \;\le\; \sqrt{6}\,\Delta t\, \max\!\left(\frac{1}{m},\ \frac{6}{\epsilon_u\,\epsilon_a},\ \frac{a_3\, l}{J_{xx}},\ \frac{a_7\, l}{J_{yy}},\ \frac{1}{J_{zz}}\right) \;<\; \infty. \tag{96}$$

The successful-landing trials of Figure 5 empirically satisfy Equation 95: the applied effort norm sits at the allocation bound during the reaching transient and then relaxes toward hover trim, never approaching zero thrust. We also note that the envelope is needed only to bound the cross-coupling *magnitude*, not the anti-windup *direction* on the thrust channel: substituting $D_{1,1} = \Delta t\ \mathrm{c}\,\phi_k\ \mathrm{c}\,\theta_k/m$ into the wrapped thrust law cancels the attitude factors, giving $\partial u_T/\partial \xi_T = -\eta_z\, \Delta t$, so a positive thrust deficit reduces the commanded thrust by the same amount at every attitude.

G. Reduction to the Unconstrained Limit

If the dSMC command u_k lies inside $\mathcal{U}$ for all k, then the saturation collapses and the controller simply returns $\bar{u}_k = u_k$ and $\Delta u_k = 0$, which is the unconstrained dSMC control effort. The auxiliary state update becomes $\xi_{k+1} = (I - K_\xi)\,\xi_k$, and so, decays to 0 exponentially as well. Equation 51 thus gives $\tilde{s}_k \to s_k$ exponentially, and the control law converges to the baseline dSMC. As $\xi_k \to 0$, the Lyapunov function reduces to

$$V_k \;\longrightarrow\; s_k^T W_s s_k. \tag{97}$$

With $W_s = \frac{1}{2} I$ this is $\frac{1}{2}\sum_\alpha s_{\alpha,k}^2$, the per-channel sum of $V_{i,k}$ from Equation 28, and clearly still positive definite in s. So, we recover our unconstrained Lyapunov positivity when the saturation ceases, as expected.

H. Steady-State Residuals Under Sustained Saturation

The steady-state response to a sustained and constant control deficit $\Delta u_k = c$ is the fixed point X^∞ of Equation 75, $(I - \mathcal{A})\,X^\infty = \mathcal{B}\,c$, treating D_k as constant ($D_k = D$) over the steady-state window. When the ideal control effort eventually returns to the feasible set and $\Delta u_k \to 0$, both ξ and $\tilde{s}$ decay to zero, and the controller resumes nominal sliding. However, under a sustained $|c| > 0$, the controller fails to converge to the sliding surface, and instead maintains some residual error, with $V^\infty \le \gamma\,|c|^2/\nu$ from the ISS bound (Equation 91), quadratic in the deficit magnitude.

The fixed-point equation

$$\begin{bmatrix} E & -D\,K_\xi \\ 0 & K_\xi \end{bmatrix} \begin{bmatrix} \tilde{s}^\infty \\ \xi^\infty \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix} \tag{98}$$

gives, from the bottom up,

$$\xi^\infty = K_\xi^{-1}\,c, \qquad \tilde{s}^\infty = E^{-1}\,D\,c. \tag{99}$$

Under matched gains (Equation 52), $K_\xi = E$ and

$$\xi_\alpha^\infty = \frac{c_\alpha}{\eta_\alpha\,\Delta t}, \qquad \tilde{s}_i^\infty = \frac{(D)_{ij}\,c_j}{\eta_i\,\Delta t}. \tag{100}$$

For a deficit on channel α, $c = c_\alpha\,e_\alpha$, the same-channel ratio $|\tilde{s}_i^\infty|/|c_\alpha| = |D_{i\,\alpha}|/(\eta_i \Delta t)$ becomes, substituting from Equation 72,

$$\frac{|\tilde{s}_z^\infty|}{|c_T|} = \frac{\mathrm{c}\,\phi\;\mathrm{c}\,\theta}{m\,\eta_T} \approx \frac{1}{m\,\eta_T}, \qquad \frac{|\tilde{s}_\phi^\infty|}{|c_{M_x}|} = \frac{a_3\,l}{J_{xx}\,\eta_{M_x}}, \tag{101}$$

$$\frac{|\tilde{s}_\theta^\infty|}{|c_{M_y}|} = \frac{a_7\,l}{J_{yy}\,\eta_{M_y}}, \qquad \frac{|\tilde{s}_\psi^\infty|}{|c_{M_z}|} = \frac{1}{J_{zz}\,\eta_{M_z}}, \tag{102}$$

which are Δt-independent. A pure thrust deficit $c = c_T\,e_1$ simultaneously perturbs the underactuated channels via column 1 of D:

$$\frac{|\tilde{s}_\phi^\infty|}{|c_T|} = \frac{|a_1|\,|g_{1,k}|}{\eta_{M_x}}, \qquad \frac{|\tilde{s}_\theta^\infty|}{|c_T|} = \frac{|a_5|\,|g_{2,k}|}{\eta_{M_y}}. \tag{103}$$

These same-channel ratios are monotone decreasing in η_α, so their infimum over the Schur-stable interval $\eta_\alpha \Delta t \in (0, 2)$ (equivalently, the discrete-time sliding-mode existence condition of Sarpturk et al. [15]) is approached only as $\eta_\alpha \Delta t \to 2$ (the Schur boundary, where the decay rate $\eta_\alpha \Delta t\,(2 - \eta_\alpha \Delta t)$ vanishes), giving $\Delta t/(2m)$ for thrust. At the decay-rate-optimal operating point $\eta_\alpha \Delta t = 1$ the ratio is instead $\Delta t/m$ for thrust and $a_3 l \Delta t/J_{xx}$ for roll (identically for pitch; $\Delta t/J_{zz}$ for yaw), so the gain choice trades residual size against convergence speed. The moment-vs-thrust residual ratio scales as

$$\frac{|\tilde{s}_\phi^\infty|/|c_{M_x}|}{|\tilde{s}_z^\infty|/|c_T|} = \frac{m\,a_3\,l}{J_{xx}} \cdot \frac{\eta_T}{\eta_{M_x}} = \frac{a_3}{l} \cdot \frac{m\,l^2}{J_{xx}} \cdot \frac{\eta_T}{\eta_{M_x}}, \tag{104}$$

in which the vehicle dependence is carried by the dimensionless mass-distribution factor $m\,l^2/J_{xx}$, equal to 17.8 for this vehicle. The ratio is therefore large whenever $m\,l^2 \gg J_{\alpha\alpha}$, the thin rotor-plane mass distribution typical of small quadrotors, which is a trait of the underactuated system and not of the saturation handler. Recovery to the sliding surface, therefore, requires the saturation deficit itself to eventually disappear; the controller alone cannot drive the residual to zero while the actuator remains saturated, as the system is unable to execute the optimal dSMC control efforts.

I. Saturated Control Results

We tested the performance of the saturated control law, against the baseline dSMC with a simple per-rotor speed clip. When saturation was enabled, we ran the priority-weighted projection (Equations 46–47) and auxiliary update steps (Equation 50); when saturation was disabled, we left the unconstrained dSMC command alone except for the per-rotor mixer clip in order to respect the physical control constraints of the vehicle. To test different deployment profiles, we tested landing points directly under the ballistic track, as well as at 100 meters and 200 meters to either side of the track. Eight deployment points are sampled uniformly at 20–80% of the trajectory arc length; each landing target (the ground projection of a deployment point plus the cross-track offset) is attempted from the neighboring deployment points at index offsets $\{-1, 0, +1, +2\}$, giving 140 valid trials of the $8 \times 5 \times 4$ grid after boundary exclusion, each with a 60 s stabilization window. We evaluate successful landings according to the criterion in Equation 40.

Figure 4 shows the per-target landing envelope under saturation. Targets across most of the central band are reached by at least one neighboring deploy; the misses concentrate at the short-range and lateral edges of the swept footprint. Figure 5 shows why the controller stays inside this envelope, as the applied control-effort norm (the post-allocation command $\bar{u}_k$) sits at the priority-allocation bound ($|\bar{u}_k| \approx 40 = 4b\,\Omega_{\max}^2$) for the first 1-2 s of the reaching transient, then relaxes toward hover trim. As discussed in VI.F and VI.H, prolonged control saturation invalidates our stability certificate, which, in our simulation, results in the system crashing. Under the unconstrained controller, the same norm spikes above 800 in that window (median per-trial peak $\approx 1.2 \times 10^3$, worst case $\approx 10^6$); under naive per-rotor clipping the norm is capped at the same bound of 40, but without the priority structure or the anti-windup state the vehicle destabilizes, and only 11 of 140 clipped trials land, versus 46 of 140 under the saturation handler (Table 7). Figure 6 aggregates the per-target data into a continuous reachability surface and locates the power-weighted centroid of reachability, which is essentially the most reachable point in the space. Table 7 collects the four landing statistics: integrated reachability, peak success ratio, half-success radius (radius about the centroid containing half the successful trials), and the centroid location, for the saturation handler enabled vs. the clipped baseline.

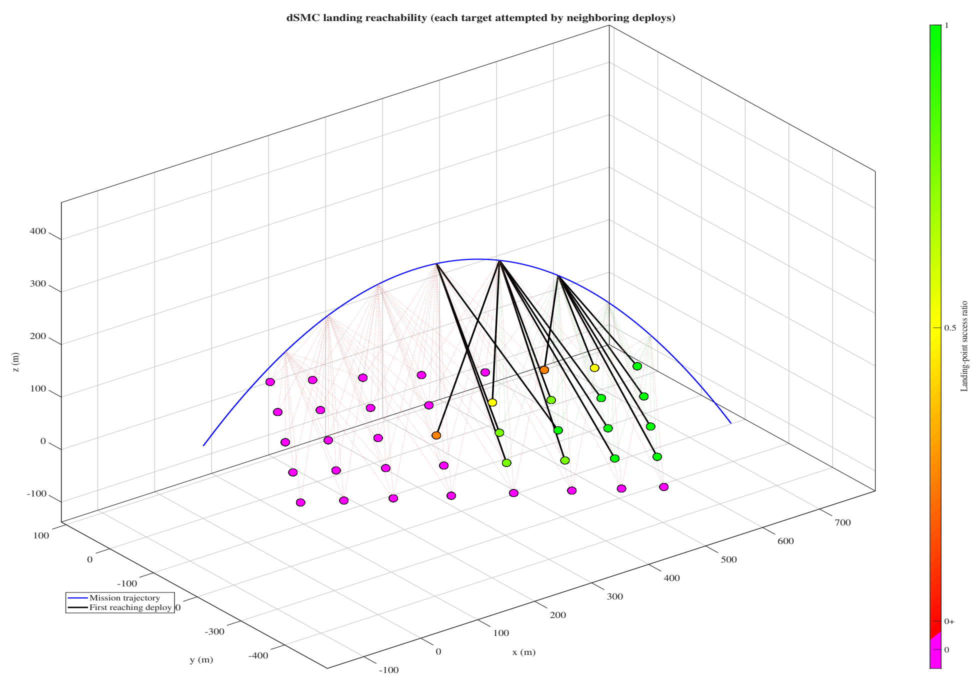

Fig. 4 Per-target landing reachability under saturation. Each landing target is attempted by neighboring deploys; markers are colored by the per-target success ratio, and black lines connect each target to its earliest reaching deploy.

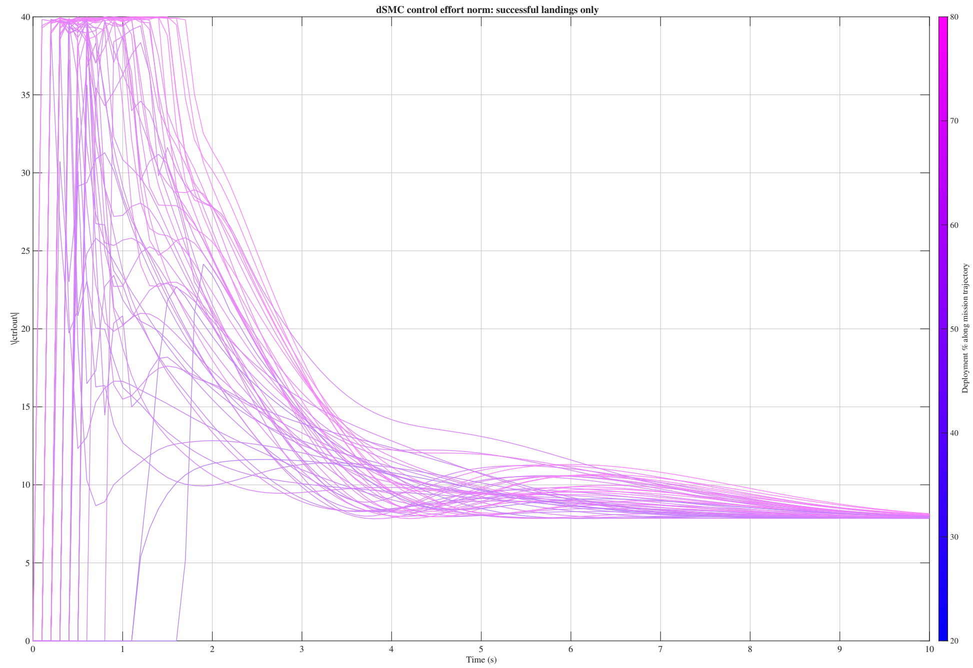

Fig. 5 Per-trial applied-command norm $|\bar{u}_k|$ during ballistic recovery with the saturation handler enabled, restricted to trials that achieved a successful landing. Traces are colored by deploy point along the mission trajectory.

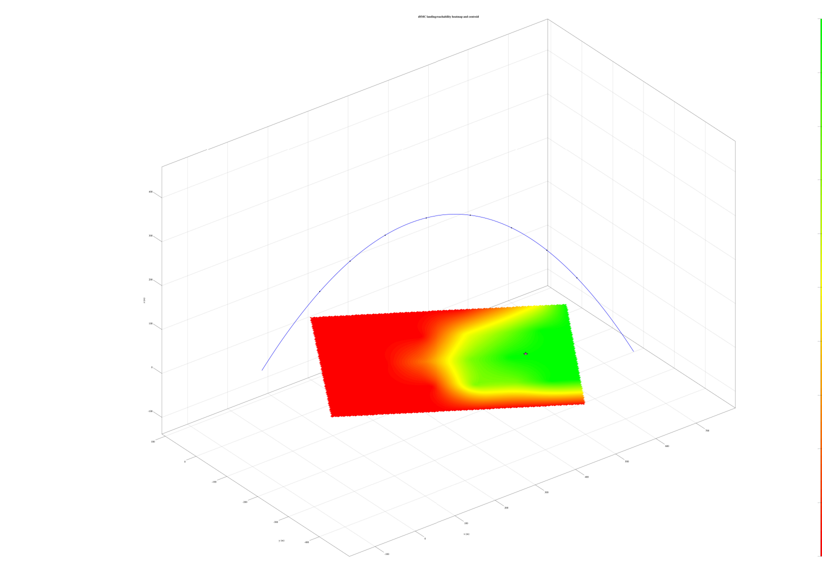

Fig. 6 Landing-success ratio over the deploy $\times$ cross-track sweep with the saturation handler enabled. The interpolated reachability surface is plotted beneath the ballistic mission trajectory, along with the power-weighted reachability centroid ($p = 2$, star).

Table 7 Landing-reachability statistics for the ballistic-recovery sweep with the saturation handler enabled versus the clipped baseline. Centroid uses the power-weighted estimator with $p = 2$.

Metric	Saturation	Control Clipping Only
Reachability fraction	0.329	0.079
Peak success ratio	0.978	0.292
Half-success radius (m)	193.24	156.19
Centroid (x, y) (m)	$(453.04, -267.16)$	$(430.53, -252.59)$

VII. Conclusions

When we assume no control saturation, dSMC can stabilize the drone at every point along the ballistic trajectory. However, when we apply basic motor speed clipping, this breaks down and requires saturation to recover performance. This matters for launched-UAV operations because, given the smaller actuators of a launched UAV, control effort constraints can be severe. The COTS techniques (LQR and PID) and the continuous sliding mode controller do not stabilize the vehicle at most points along the trajectory - this is consistent with our intuition, as the COTS controllers assume linearity and the cSMC controller assumes continuity. For launched-drone operations, or for off-nominal initial conditions and aggressive flight in general, the dSMC formulation here beats the COTS controllers and, unlike cSMC, keeps its stability certificates under our discretized control plant.

A. Future Work

With an effective saturated controller and centroid generation logic in place, we want to additionally study the inverse problem of mapping from a desired centroid to the ballistic targeting variables (launch azimuth, elevation, velocity, exit spin, etc) required for a min-time landing, a max-likelihood landing, and a min-effort landing. Currently, we are examining numerical surrogate-based approaches to solving this, and have working prototypes in simulation. We plan to move to flight-test verification after, focusing on the ballistic case. Measuring real moments of inertia and drag parameters is the first step toward closing the sim-to-real gap. Inclement-weather operations will require a basic weather model. However, the stability certificates of both dSMC demonstrated on the ballistic envelope suggests that minor velocity and orientation perturbations won't destabilize the controller.

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Generative AI Statement

The Anthropic Claude [19] generative AI tool was used to improve the formatting and layout of the equations and tables, to reformat Microsoft Word equations to LaTeX formatting, to convert from MATLAB code to LaTeX equations, and to improve general syntax and grammar. The author assumes full responsibility for all results and derivations presented in this paper, and any errors which may be present therein.

References

[1] Bouman, A., Nadan, P., Anderson, M., Pastor, D., Izraelevitz, J., Burdick, J., and Kennedy, B., "Design and Autonomous Stabilization of a Ballistically-Launched Multirotor," *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 8511–8517. https://doi.org/10.1109/ICRA40945.2020.9197542.

[2] Denton, H., Benedict, M., and Kang, H., "System identification of a thrust-vectoring, coaxial-rotor-based gun-launched micro air vehicle in hovering flight," *International Journal of Micro Air Vehicles*, Vol. 17, 2025. https://doi.org/10.1177/17568293251361078.

[3] Harper, J., "Army's Fiscal 2026 Budget Proposal Aims to Equip Infantry Brigades with More Kamikaze Drones," , Jun. 2025. URL https://defensescoop.com/2025/06/27/army-fiscal-2026-budget-request-loitering-munitions-drones-lasso/.

[4] Starks, M. A., Chu, J., Leung, K. H. B., Blewer, A. L., Simmons, D., Hansen, C. M., Joiner, A., Cabañas, J. G., Harmody,

M. R., Nelson, R. D., McNally, B. F., Ornato, J. P., Granger, C. B., Chan, T. C. Y., and Mark, D. B., "Combinations of First Responder and Drone Delivery to Achieve 5-Minute AED Deployment in OHCA," *JACC: Advances*, Vol. 3, No. 7, Part 2, 2024, p. 101033. https://doi.org/10.1016/j.jacadv.2024.101033.

[5] Liang, M., and Delahaye, D., "Drone Fleet Deployment Strategy for Large Scale Agriculture and Forestry Surveying," *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019, pp. 4495–4500. https://doi.org/10.1109/ITSC.2019.8917235.

[6] Michelena, T., "Swarm Technology in Sustainment Operations," , Jan. 2025. URL https://www.army.mil/article/282467/swarm_technology_in_sustainment_operations.

[7] Xiong, J.-J., and Zhang, G., "Discrete-time sliding mode control for a quadrotor UAV," *Optik*, Vol. 127, No. 8, 2016, pp. 3718–3722. https://doi.org/10.1016/j.ijleo.2016.01.010.

[8] Xu, G., Xia, Y., Zhai, D.-H., and Ma, D., "Adaptive Prescribed Performance Terminal Sliding Mode Attitude Control for Quadrotor Under Input Saturation," *IET Control Theory & Applications*, Vol. 14, No. 17, 2020, pp. 2473–2480. https://doi.org/10.1049/iet-cta.2019.0488.

[9] Kuang, J., and Chen, M., "Adaptive Sliding Mode Control for Trajectory Tracking of Quadrotor Unmanned Aerial Vehicles Under Input Saturation and Disturbances," *Drones*, Vol. 8, No. 11, 2024, p. 614. https://doi.org/10.3390/drones8110614.

[10] Shao, X., Sun, G., Yao, W., Liu, J., and Wu, L., "Adaptive Sliding Mode Control for Quadrotor UAVs With Input Saturation," *IEEE/ASME Transactions on Mechatronics*, Vol. 27, No. 3, 2022, pp. 1498–1509. https://doi.org/10.1109/TMECH.2021.3094575.

[11] Faessler, M., Falanga, D., and Scaramuzza, D., "Thrust Mixing, Saturation, and Body-Rate Control for Accurate Aggressive Quadrotor Flight," *IEEE Robotics and Automation Letters*, Vol. 2, No. 2, 2017, pp. 476–482. https://doi.org/10.1109/LRA.2016.2640362.

[12] Mueller, M. W., and D'Andrea, R., "A Model Predictive Controller for Quadrocopter State Interception," *2013 European Control Conference (ECC)*, Zürich, Switzerland, 2013, pp. 1383–1389. https://doi.org/10.23919/ECC.2013.6669415.

[13] Wang, Y., Cao, Y.-Y., and Sun, Y.-X., "Stability Analysis and Anti-Windup Design for Discrete-Time Systems by a Saturation-Dependent Lyapunov Function Approach," *Proceedings of the 16th IFAC World Congress*, Vol. 38, 2005, pp. 759–764. https://doi.org/10.3182/20050703-6-CZ-1902.00782.

[14] Jiang, Z.-P., and Wang, Y., "Input-to-State Stability for Discrete-Time Nonlinear Systems," *Automatica*, Vol. 37, No. 6, 2001, pp. 857–869. https://doi.org/10.1016/S0005-1098(01)00028-0.

[15] Sarpturk, S. Z., Istefanopulos, Y., and Kaynak, O., "On the Stability of Discrete-Time Sliding Mode Control Systems," *IEEE Transactions on Automatic Control*, Vol. 32, No. 10, 1987, pp. 930–932. https://doi.org/10.1109/TAC.1987.1104468.

[16] Tahir, A., Böling, J. M., Haghbayan, M.-H., and Plosila, J., "Comparison of Linear and Nonlinear Methods for Distributed Control of a Hierarchical Formation of UAVs," *IEEE Access*, Vol. 8, 2020, pp. 95667–95680. https://doi.org/10.1109/ACCESS.2020.2988773.

[17] McCoy, R. L., *Modern Exterior Ballistics: The Launch and Flight Dynamics of Symmetric Projectiles*, Schiffer Publishing, Limited, 1999. URL https://books.google.com/books?id=ay5UPQAACAAJ.

[18] Pye, J. E., "Controller Analysis Simulation Environment," , 2025. URL https://github.com/jepye-3141/controller_analysis#.

[19] Anthropic, "Claude Opus 4," , 2025. URL https://www.claude.ai.